PREDICTIVE NATURAL LANGUAGE METRICS OF ALZHEIMER'S DISEASE AND COGNITIVE DECLINE TREND ANALYSIS

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A Thesis Presented to the Graduate Faculty of the

Lyle School of Engineering

Southern Methodist University

in

Partial Fulfillment of the Requirements

for the degree of

Master of Science

with a

Major in Computer Science

by

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May 16th, 2026

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Predictive Natural Language Metrics of Alzheimer's Disease and Cognitive Decline Trend Analysis

Advisor: Dr. Jennifer Dworak

Master of Science conferred May 16th, 2026

Thesis completed April 13th, 2026

This thesis, inspired by the historic and recent work on the Nun Study [1] founded and first directed by Dr. David Snowden, a continuing longitudinal study on volunteers from the School Sisters of Notre Dame congregation, aims to investigate predictive metrics of Alzheimer's disease for its diagnosis, based on patients' linguistic ability quantified by Propositional Idea Density (PID) of the patient's writing. In this work, we thoroughly investigate traditional natural language processing metrics (NLP) as well as the more modern NLP techniques based on training Artificial Neuro Networks (ANNs) through machine learning (ML). Furthermore, we apply capable linguistic ability metrics to certain datasets to evaluate potential cognitive decline of a certain population. We find in recent years, internet users' comments of popular websites and scholarly peer-review comments of prestigious research conferences and journals has seen year-to-year decline for certain domains, yet also incline in others. This work explores NLP metrics developed by ML algorithms and discovers the clear overall trends of PID of certain domains of user comments, and suggests that linguistic ability metrics should be applied on longitudinal text datasets to discover trends.

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Thanks for the extensive support of my committee members, Dr. Dworak, Dr. Zhang and Dr. Larson, for their advising, support and encouragement on my work.

Chapter 1

Introduction

This thesis aims to serve as the graduation paper submitted to the Computer Science and Engineering Theses and Dissertations section of the SMU Scholar digital research repository, to serve as the graduation paper for my Master of Science degree in Computer Science. This thesis is advised by Dr. Jennifer Dworak, Professor in the Department of Electrical and Computer Engineering at the Lyle School of Engineering, who is a professor of Computer Science by Courtesy. Dr. Dworak chairs the committee as the thesis advisor. Dr. Jia Zhang and Dr. Eric Larson, both Professors in the Department of Computer Science, serve as committee members.

This thesis, inspired by the historic and recent work on the Nun Study [1] founded and first directed by Dr. David Snowden, a continuing longitudinal study on volunteers from the School Sisters of Notre Dame congregation, aims to investigate predictive metrics of Alzheimer's disease (AD) for its diagnosis based on patients' linguistic ability such as idea density of the patient's writing. Dr. David Snowden's work popularized the use of Propositional Idea Density (PID) as a metric for analyzing dementia patients' linguistic ability, due to the statistical association of lower PID values in patient writings and higher risk and symptoms in AD. Furthermore, applying the linguistic ability metric PID to certain datasets can evaluate potential cognitive decline of a certain population. We find in recent years, internet users' comments of popular websites and scholarly peer-review comments of prestigious research conferences and journals have seen year-to-year decline for certain domains, yet also increase in others. This work explores NLP metrics developed by ML algorithms and discovers the clear overall trends of PID of certain domains of user comments, and suggests that linguistic ability metrics should be applied on longitudinal text datasets to discover trends.

In this paper, we investigate the transfer learning ability of encoder language models

in predicting if the patient speech transcription comes from a patient with dementia, what contributes to their prediction of dementia transcripts, and compare their ability to predict with PID. Furthermore, we investigate the overall longitudinal trend of linguistic ability decline over the years of various internet fora, social media sites and scientific conferences and journals.

This work justifies the use of PID to investigate linguistic ability trend for a specific population on a specific domain of topics, as we see steady and statistically significant increase and decline of linguistic ability over various domains. Future work should focus on the original Nun Study for the analysis of more formal writing style rather than verbal tasks, as well as developing a more consistent metric to improve evaluation of linguistic ability for trend analysis of domain-specific text data.

Chapter 2

Related Works

2.1 The Nun Study

The Nun Study is a “longitudinal study of 678 Catholic sisters 75 to 107 years of age who are members of the School Sisters of Notre Dame congregation” [4]. Each volunteer has agreed to annual check-ups and the release of certain documents and bio-medical records that helped Dr. Snowden’s studies, including autobiographies written upon admission to the school at mean age of 22 years [5].

The evaluation metric that correlates to later-life cognitive decline has proven to be idea density of personal writings. “Idea density scores from early life had strong inverse correlations with the severity of Alzheimer’s disease” [6]. Though the exact statistical method is not given, Snowden has proven that “Correlations between idea density scores and neurofibrillary tangle counts were -0.59 for the frontal lobe, -0.48 for the temporal lobe, and -0.49 for the parietal lobe (all p values < 0.0001, which indicates a strong statistical significance). Idea density scores were unrelated to the severity of atherosclerosis of the major arteries at the base of the brain and to the presence of lacunar and large brain infarcts. Low linguistic ability in early life may reflect suboptimal neurological and cognitive development, which might increase susceptibility to the development of Alzheimer’s disease pathology in late life.” This urges work on finding the correlations between the development of AD and the suboptimal cognitive development demonstrated in early life linguistic ability.

2.2 Propositional Idea Density (PID)

Propositional Idea Density (PID) was defined as "the average number of ideas expressed per ten words for the last ten sentences of each autobiography." [6] This serves as a natural language processing technique that evaluates a patient’s linguistic ability as a deterministic score. To process the natural language, ideas are defined as such: "Ideas corresponded to

elementary propositions, typically a verb, adjective, adverb, or prepositional phrase." [6] The intuition is that more frequent ideas occur in a patient's speech or writing, more able is the patient in linguistic ability.

Dr. Snowden's work demonstrates that PID is a reliable metric for analyzing patients' linguistic ability. After Snowden's discovery, many publications also adopted PID as their metric to quantify linguistic ability [7] [8] [9], with many using it as a clinical metric. [10] [11] Authors of these related works have all used PID as a metric since "propositional idea density (PID) is associated with an increased risk of developing AD." [11] With precedents, we will also use PID to evaluate linguistic ability, suggesting the increase of which is strongly associated with increased risk of developing AD for a certain population.

Since PID is a widely used metric in clinical linguistics to quantify the rate at which new information is expressed in a text relative to its length, in the context of AD research, PID serves as a proxy for semantic processing efficiency. This section discusses two primary computational approaches to measuring PID: Part-of-Speech-based Computerized Propositional Idea Density Rater (CPIDR) [12] developed according to the metric used by Dr. Snowden in his initial findings and the Dependency-based Propositional Idea Density (DEPID). [8]

2.2.1 Computerized Propositional Idea Density Rater (CPIDR)

The Computerized Propositional Idea Density Rater (CPIDR), introduced by Brown et al., was designed to automate the manual propositional analysis techniques. CPIDR calculates PID by identifying specific parts of speech (POS) that typically carry semantic weight, such as verbs, adjectives, adverbs, and prepositions.

The core calculation for Propositional (idea) density (PD) in CPIDR is defined as:

$$PD = \frac{\text{Count of Propositions}}{\text{Total Number of Words}} \tag{2.1}$$

To refine this count, CPIDR utilizes the MontyLingua tagger and applies a series of adjustment rules to avoid over-counting auxiliary structures. The adjusted proposition count is modeled as:

$$\text{Propositions} = V + Adj + Adv + Prep + Conj + Det_{other} + M_{neg} - V_{aux} - V_{link} \tag{2.2}$$

where V, Adj, Adv, $Prep$, and $Conj$ represent verbs, adjectives, adverbs, prepositions, and conjunctions, respectively. Deductions are made for auxiliary verbs (V_{aux}) and linking verbs (V_{link}) to ensure that complex verb phrases are treated as single semantic units.

2.2.2 Dependency-based Propositional Idea Density (DEPID)

While POS-based methods are computationally efficient, they often fail to capture the underlying structural relationships between words. To address this, Sirts et al. proposed DEPID [8], which utilizes dependency parsing to identify propositions based on semantic relations (e.g., predications, modifications, and connections) rather than isolated word classes.

The DEPID score is calculated as the ratio of qualifying dependency relations to the total word count (N):

$$DEPID = \frac{|\mathcal{P}_{dep}|}{N} \tag{2.3}$$

A significant contribution of this framework to AD research is the introduction of *DEPID-R*. Patients with dementia often exhibit perseveration, aka the frequent repetition of ideas. DEPID-R accounts for this by only counting unique dependency relations, thereby providing a measure of information novelty:

$$DEPID\text{-}R = \frac{|\text{unique}(\mathcal{P}_{dep})|}{N} \tag{2.4}$$

By filtering out redundant semantic structures and utilizing token-level filters for vague determiners and empty subjects, DEPID provides a more robust metric for analyzing the fragmented speech patterns typical of the Pitt Corpus and similar clinical datasets.

Furthermore, due to the different approaches CPIDR and DEPID take while calculating idea density scores, if the CPIDR and DEPID scores diverges, we can gain critical insight into the relationship between lexical volume and structural substance. A linguistic profile characterized by high CPIDR but low DEPID typically indicates inflated density due to lexical repetition/perseveration; while the speaker utilizes a high frequency of content parts of speech, these words fail to form novel or complex structural relationships. This phenomenon is frequently observed in AD-related speech, where patients may maintain the outward appearance of content through repetitive descriptors or fragmented phrases that

CPIDR fails to penalize. Conversely, a low CPIDR relative to a high DEPID suggests syntactic-semantic complexity, representing a concise communication style where information is packed efficiently into complex syntactic links. Ultimately, analyzing the discrepancy between these two metrics allows researchers to distinguish between the mere presence of propositional word classes and the successful construction of unique semantic ideas, offering a more nuanced diagnostic window into cognitive decline.

2.3 Artificial Neural Networks (ANNs) as Natural Language Processing (NLP) Techniques

With the recent advancement of Artificial Neural Networks (ANNs), there have been applicable and replicable precedents of research that predicts AD with biomedical metrics. A more recent work by Snowden's team found that training an ANN over patients' neuropathological features, including lesion amounts in different areas of the brain, AD can be perfectly predicted within the controlled population of the Nun Study. "By taking into account the above four neuropathological features, the overall predictive capability of ANNs in sorting out AD cases from normal controls reached 100%" [13].

More recently, with the advancement of Transformer architecture [14] applied in Natural Language Processing (NLP), text embedding models like Bidirectional Encoder Representations from Transformers (BERT) [15] have been widely implemented in the field of AI for health. A group of researchers "developed a deep learning framework using BERT models which provide an effective solution for prediction of MCI-to-AD progression using clinical note analysis" [16]. By finetuning BERT encoding large language models with segmented clinical notes of patients, the model predicts MCI-to-AD risk with marginally better results than other BERT models. However, this model aims to predict patient cognitive decline based on clinical notes from doctors' evaluations. Instead, we can aim to evaluate patient transcript for cognitive decline potentials. In a recent work where I helped to finetune RoBERTa model on patient transcripts for personality predictions, the CLS embeddings show clear correlations between the embedding of the text and the multi-dimensional value of the personality trait trained upon [17]. This work demonstrates the capability of BERT embeddings on patient transcripts to evaluate certain linguistic metrics, where it may be more broadly used on diverse patients to predict their risk in AD and cognitive decline.

Chapter 3

Research Goals

This thesis will explore the following two research questions.

3.1 Investigating ANNs' Potential to Predict Cognitive Decline within the Pitt Corpus

Research question: can Transformer-based ANNs in modern Machine Learning (ML) as NLP techniques attend to linguistic patterns that may be associated with an individual's cognitive decline?

The first goal aims to investigate if Transformer-based ANNs, such as BERT encoding language models, are able to learn what linguistic attributes correlate with cognitive decline. We will experiment with various NLP techniques, by training and fine-tuning ML models on a specific dataset, to analyze if ANN techniques can predict if a speech comes from a cognitive-declining individual.

3.2 Analyzing Longitudinal Cognitive Ability Trend over Corpora of Datasets

Research Question: can we demonstrate a linguistic ability trend for a specific population of writers based on the PID score of their comments?

The second goal will analyze seven corpora of data:

- Reddit Comments (sampled datasets)
- Common Crawl (sampled datasets)
- Airliners.net All Fora Comments
- Hacker News All Fora Comments
- TheForce.net All Fora Comments

- The International Conference on Learning Representations (ICLR) Peer-Review Comments
- BioMed Central Journal (BMC) Peer-Review Comments

These datasets will give us a wide range of domains of text, across global demographics, user bases (internet and academia), discussions (comments and posts) and intents of comments. The oldest forums trace back to 1998 (Airliners.net and TheForce.net), and have a consistent amount of data till present day (cut-off date: March 2026). This will provide the proper insights to the linguistic ability trend across each domain of writers based on the PID scores of their comments in the past up to 29 years, providing us insights into what is the linguistic ability trend across these different populations of writers and how does the domain writing PID score vary over time.

Chapter 4

Investigate ANNs' Potential to Predict Cognitive Decline

4.1 Methodology

Since the desired data from the Nun Study was not available during the work of my thesis, we substituted the analysis of the Nun Study early life autobiography writings with an available dataset, the Pitt Corpus from DementiaBank. [18] The Pitt Corpus encompasses transcribed interviews of control subjects and individuals diagnosed with Dementia, categorized into three distinct task types: (1) Cookie, consisting of responses to the Cookie Theft stimulus photo from both groups; (2) Fluency, involving word fluency tasks for the Dementia group only; and (3) Recall, featuring story recall tasks for the Dementia group, where the category utilizes two narratives, including the "George and Melanie" story [19], which describes a grandfather's visit to the city with his granddaughter.

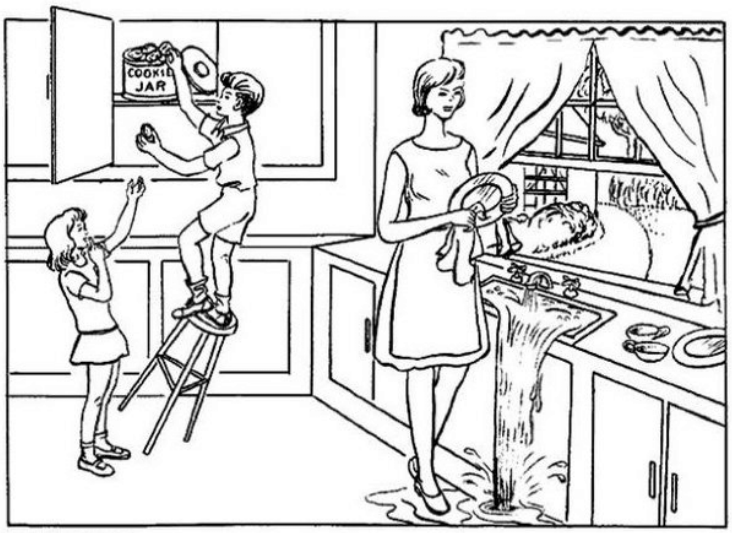

Figure 4.1: The Cookie Theft picture. Participants were asked to describe what they see in the image. [2]

4.1.1 Data Preprocessing

The dataset consists of 293 transcripts (195 Dementia, 98 Control), presenting a class imbalance. Data sanitization involved removing transcription artifacts, such as the ASCII Negative Acknowledge (`\x15`) character, to prevent models from learning spurious correlations due to the lack of tokenization of these special characters in embedding language models such BERT, leaving just pure natural language texts. We employed a stratified 5-fold cross-validation scheme during training to preserve the class distribution across training and validation splits.

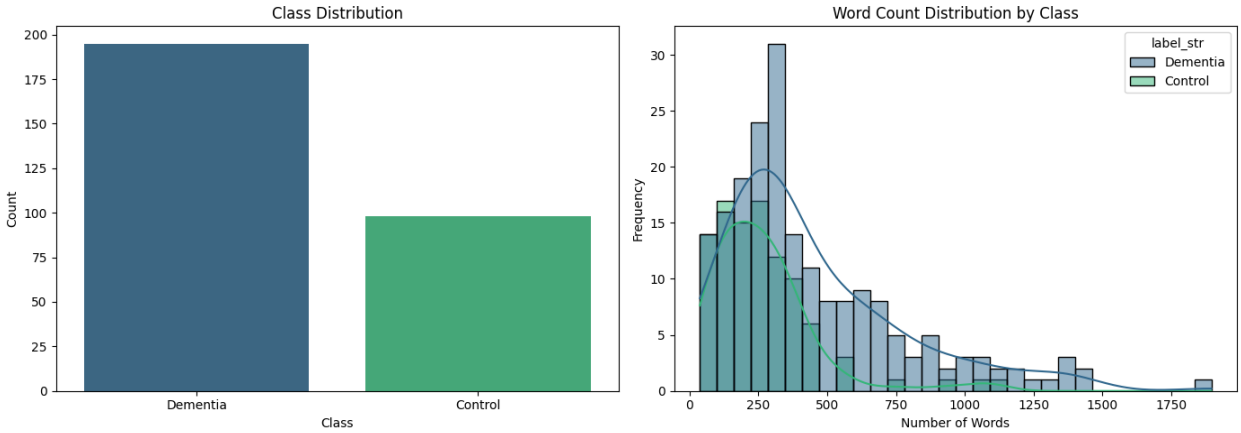

Figure 4.2: The class distribution and word count distribution of the Pitt Corpus.

Regarding the class imbalance of the dataset, as there are 87 more dementia-labeled data points than control-labeled data points, no data point is imputed or removed. We refrained from imputing or removing data because the linguistic variability of the transcripts is unique to each participant. Moreover, removing 87 samples from the dementia cohort would lead to a substantial loss of information in an already data-scarce environment. However, with robust cross-validation for training and validation splits, as well as monitoring the minority class and macro average precision, recall and F1 scores, which will indicate if a model predicts with bias towards the majority class. During training, we

employ the Adam optimizer [20], which stabilizes training with momentum-aware adaptive learning rates that account for imbalanced gradients and prevent slow learning on minority classes, as well as weighted binary cross-entropy loss with class weights computed via sklearn's `compute_class_weight('balanced')` (yielding weights of `[1.50, 0.75]` for Control and Dementia respectively) to account for the data class imbalance. All transcripts are lowercased prior to tokenization, as conversational speech transcripts contain no meaningful casing distinctions.

4.1.2 Linguistic Rule-Based Baselines

To establish a baseline using established linguistic theories, we computed Computerized Propositional Idea Density (CPIDR) [12] and Dependency-based Propositional Idea Density (DEPID) [8] directly from the text. DEPID-R is also analyzed, which like DEPID, is also a linguistic structural dependency-based propositional idea-density metric, yet also being able to detect and exclude idea repetitions, a common symptom of dementia patients. We conducted an independent statistical analysis, point-biserial correlation and independent T-tests, to evaluate their predictive power independent of machine learning models.

4.1.3 Machine Learning and Semantic/Pragmatic ANN Architectures

To push beyond traditional linguistic metrics, we iteratively developed and evaluated multiple machine learning architectures:

1. **TF-IDF Baselines**: To enable learning on the vocabulary patterns in our dataset, we established non-linear ANNs baselines utilizing term frequency-inverse document frequency (TF-IDF) [21] feature extraction over the top 5,000 unigrams and 5,000 trigrams. TF-IDF examines a word's relevance to a specific document within the entire dataset by calculating the term frequency (TF) that measures how frequently a word appears in a transcript, multiplied by the inverse document frequency (IDF) that measures how rare a word is across all transcripts in our dataset. We train two models, one with the most frequent 5,000 unigrams (one word) of the datasets and another model with most frequent 5,000 trigrams (a triplet of adjacent words) to train the model on the correlation between the unigram or trigram and if the patient has dementia based on the individual's transcript. TF-IDF is used commonly in NLP

tasks since it enables simple ANNs to take word frequency within a document against its rarity across a dataset as input, to learn meaningful words that contribute to the label prediction while de-emphasizing commonly occurring vocabulary such as stop words. The TF-IDF sparse matrix that stores the importance of words across the entire datasets was then mapped to:

- *XGBoost*: A gradient boosted tree classifier [22] configured with `n_estimators=100`, `max_depth=3`, an aggressive `learning_rate=0.05`, L_1 regularization `alpha=1.0`, and `scale_pos_weight=0.5` (the ratio of Control to Dementia samples in the training set) to counterbalance the Dementia class majority. XGBoost is particularly effective when the TF-IDF matrix is sparse, as the model learns by sequentially constructing decision trees where each new tree reduces the residual error of the previous ensemble.
- *Multi-Layer Perceptron (MLP)*: A PyTorch MLP [23] with hidden layers `(64, 32)`, ReLU activations, and Dropout regularization (`p=0.1`) between layers, trained with the Adam optimizer [20] (`lr=1e-3`) and weighted binary cross-entropy loss over 100 epochs with a batch size of 32. MLP learns non-linear, complex mappings between inputs and outputs by optimizing internal weights and biases through backpropagation, with each artificial neuron computing weighted sums of neurons of previous layers to construct a non-linear decision boundary.

2. **Sequential Models**: To capture syntactic degradation via sequence recurrence, we converted tokens into learned 100-dimensional embeddings via a padding-aware lookup table (vocabulary size 5,000) and propagated them through temporal networks using the Adam optimizer [20] (`lr=1e-3`) with weighted binary cross-entropy loss to account for the class imbalance. Using models that utilize hidden states (latent representations) to capture dependencies across adjacent words, sequential models have the capacity to learn syntactic patterns rather than merely vocabulary frequencies.

   - *1D Convolutional Neural Network (CNN)*: A lightweight spatial CNN [24] trained over 15 epochs with batch size 32, utilizing sliding-window filters across text sequences, excelling at capturing localized syntactic fragments. Transcripts are

tokenized and mapped to continuous word embeddings via a learned embedding lookup table to a dimension of 100, which are then processed by parallel 1D convolutional layers to extract local n-gram features with 100 input channels and 128 output channels of trigram, quadrigram and pentagram kernels. The most salient features are captured via max-over-time pooling, concatenated, regularized with Dropout (`p=0.5`), and passed through a fully connected layer to produce the final diagnostic prediction. To elucidate the model's decision-making, the architecture is explicitly modularized to separate the continuous embedding space, enabling the application of Integrated Gradients [25] that allows gradient-based attribution scores to be calculated across the embeddings and aggregated dimensionally, directly quantifying and visualizing each individual word's contribution to the prediction.

- *Bidirectional Long Short-Term Memory (BiLSTM)*: A BiLSTM [26] implemented with an additive Attention mechanism [27] across the backward and forward temporal states, trained over 100 epochs with batch size 32, to model concurrence between each two tokens which contextualizes long-range pausing and grammatical decay. The classification architecture employs a Bidirectional Long Short-Term Memory (BiLSTM) network augmented with a custom attention mechanism. Discrete input tokens are first mapped to 100-dimensional continuous vectors via a padding-aware lookup embedding layer. These embeddings sequentially feed into a two-layer BiLSTM with a hidden state dimension of 128 and an inter-layer dropout rate of 0.3 to mitigate overfitting; the bidirectional processing yields concatenated contextualized representations of 256 dimensions for each token in the sequence. To dynamically capture the most salient linguistic features, an attention module applies a linear projection followed by a sequence-wise softmax activation to compute normalized scalar importance weights for every timestep. The BiLSTM output states are then linearly combined according to these attention weights to form a single, focused context vector. Finally, this weighted context vector is projected through a fully connected linear layer to compute the final binary classification logits.

3. **BERT Fine-Tuning**: Bidirectional Encoder Representations from Transformers (BERT) [15] is a transformer architecture [14] that utilizes the attention mechanism [27], specifically training the model to train the query, key, value weights to calculate the attention matrix, the softmax of the scaled dot-product of query and key matrices multiplied by the value matrix in each attention head, to attend to the co-occurrence of each input tokens to each output tokens. Based on the transformers ANN architecture, Devlin et al. trained an LLM with Masked Language Modeling (predict a masked token in the middle of a sentence) and Next Sentence Prediction (predict if the second sentence is the next sentence of the first sentence) objectives. The BERT language model is pre-trained on massive, unstructured corpora of texts: English Wikipedia (2.5 billion words) and BooksCorpus (800 million words), totaling to 16 GB of text data (3.3 Billion words). BERT's transformer architecture, pre-training protocols, as well as the amount of wide domain range of texts, enabled the LLM to gain Natural Language Understanding capabilities and ability to accurately predict tokens of the Masked Language Modeling and Next Sentence Prediction objectives, that not only captures the syntactic attributes of natural language, but also the semantic and pragmatic attributes. The pre-trained BERT, upon embedding any sequence of text, will generate a CLS token that has a dimension of 768 as the first token of the final hidden layer output. The CLS token captures a summary representation of the entire text output, therefore it has become a common practice for NLP researchers to attach a classifier on top of the CLS token to enact classification tasks. [28] We used two BERT variant language models that are trained upon the pre-trained BERT language model:

   - *DistillBERT*: with model name `distilbert-base-uncased` [29], DistillBERT is pretrained by knowledge distillation from the pre-trained BERT language model by initializing a random-weight BERT architecture model with fewer parameter counts, and then trained with BERT language model as the teacher model, optimizing the student model (DistillBERT) with language modeling loss, distillation loss, cosine-distance loss so the student model will perform very similarly to the teacher model while being smaller in architecture: BERT base has a total pa-

rameter count of 110 million, while DistillBERT has a total parameter count of 66 million, having a model size of 40% while retaining over 95% of its language understanding performance. Using a smaller language model not only alleviates computation constraints (memory, GPU acceleration) but also enables the model to converge faster with less data, which can be desirable for scarce resource domain adaptation.

- *roberta-base*: with model name `roberta-base` [30], Robustly Optimized BERT Approach (roBERTa) is post-trained upon the pre-trained BERT language model by Fine-Tuning with Common Crawl News, Open Web Text, and Common Crawl Stories (144 GB of text additional to pre-trained BERT), extending the domain knowledge of BERT while retaining the model parameter counts and architecture.

We leveraged pre-trained contextual embedding LLMs with BERT architectures to train on the semantic and pragmatic linguistic features. We integrated both DistilBERT and RoBERTa through two distinct pathways:

- *Frozen Base + MLP*: We extracted the CLS token's 768-dimensional embeddings from the last hidden layer and classified them through a PyTorch MLP with hidden layers `(64, 32)`, ReLU activations, and Dropout (`p=0.1`), trained with Adam (`lr=1e-3`) and weighted binary cross-entropy loss over 100 epochs with batch size 32. All input sequences were truncated to a maximum of 512 tokens, losing a substantial amount of information on longer transcripts.
- *Full Fine-Tuning*: The entire encoder weights were updated using the HuggingFace `WeightedTrainer` (a custom `Trainer` subclass injecting balanced class-weighted binary cross-entropy loss) with AdamW (`lr=2e-5`, `weight_decay=0.01`), evaluating at each epoch and retaining the checkpoint with the lowest validation loss. Sequences were padded and truncated to 512 tokens over 15 training epochs with batch size 32.

4. **ModernBERT Fine-Tuning**: ModernBERT [31] is a recent encoder-only transformer that extends the BERT architecture with Flash Attention 2 [32] for efficient

long-context processing, rotary positional embeddings (RoPE) [33] supporting sequences up to 8,192 tokens, and alternating local-global attention layers. Pre-trained on 2 trillion tokens of diverse English text, ModernBERT-base retains a comparable parameter count to RoBERTa-base (149M vs. 125M) while achieving state-of-the-art results on standard NLU benchmarks. The 8,192-token context window is particularly relevant for our task, as many Pitt Corpus transcripts exceed RoBERTa's 512-token limit due to extended conversational tangents. We integrated ModernBERT through two pathways:

- *Frozen Base + MLP*: Static 768-dimensional CLS embeddings extracted from the full transcript (up to 8,192 tokens) and classified through the same MLP architecture as other frozen encoder experiments (hidden layers `(64, 32)`, 100 epochs, batch size 32).
- *LoRA Fine-Tuning*: Parameter-efficient LoRA adaptation ($r = 32$, $\alpha = 64$, `lora_dropout=0.1`) targeting ModernBERT's fused attention projection matrices (`Wqkv`, `Wo`). Dynamic padding via `DataCollatorWithPadding` replaced fixed-length padding to prevent memory waste on the 8,192-token maximum, and gradient checkpointing with `bfloat16` precision was enabled. The model was trained with the `WeightedTrainer` using AdamW (`lr=2e-5`, `weight_decay=0.01`) over 15 epochs with batch size 32.

5. **RoBERTa + GRU Sliding Window**: Standard BERT-family encoders impose a hard 512-token context limit: any transcript content beyond this boundary is silently truncated before the model ever processes it. For the Pitt Corpus, where many Dementia transcripts exceed 512 sub-word tokens due to extended tangential speech, repetitive phrasing, and interviewer-participant exchanges, this truncation discards precisely the late-transcript linguistic degradation that is most clinically informative. Naively increasing the context window (as ModernBERT does with its 8,192-token limit) addresses the information loss but sacrifices the localized semantic granularity that RoBERTa's 512-token attention window provides: within a 512-token span, every token attends to every other token via full quadratic self-attention, yielding dense contextualized representations. In longer contexts, the attention mechanism must dis-

tribute its capacity across thousands of tokens, diluting the per-token representational quality.

This methodology was previously done in Language-based AI modeling of personality traits and pathology from life narrative interviews by Oltmanns et al. [17], where as a co-author, I helped finetuning roBERTa for domain knowledge adaptation. Our sliding window architecture resolves this tension by preserving RoBERTa's full-attention granularity within each 512-token window while recovering the complete transcript content across windows. Each transcript is partitioned into overlapping windows (window size 512, stride 462, overlap 50 tokens), and RoBERTa independently processes each window to produce a 768-dimensional [CLS] embedding that captures the localized semantic content of that segment. The 50-token overlap ensures that no sentence-level context is severed at window boundaries: tokens near the edge of one window are re-encoded with their surrounding context in the adjacent window, preventing artificial discontinuities in the representation. The variable-length sequence of window-level [CLS] embeddings is then aggregated through a 2-layer bidirectional GRU [34] (hidden dimension 256) equipped with an additive attention mechanism. The bidirectional GRU captures temporal dependencies across sequential windows, modeling how linguistic coherence evolves, degrades, or fluctuates across the duration of the transcript, while the additive attention layer computes a learned scalar importance weight for each window, producing a single transcript-level representation as the attention-weighted sum of the GRU hidden states.

The intuition for why this hierarchical decomposition is effective for cognitive decline detection lies in the clinical phenomenology of dementia-related speech. Dementia-related linguistic degradation does not manifest uniformly across a transcript: patients may begin with relatively coherent descriptions of the Cookie Theft picture, then progressively lose propositional structure as the interview continues, or they may cycle between lucid passages and disorganized, tangential speech interspersed with fillers and repetitions. A single 512-token truncated encoding compresses this heterogeneous signal into one vector, losing the temporal trajectory of degradation. By contrast, our

sliding window approach preserves the full temporal structure: each window's [CLS] captures a local "snapshot" of linguistic quality, and the GRU models how these snapshots evolve sequentially. The attention mechanism then learns which snapshots are most discriminative, effectively implementing a soft selection over transcript regions. For a Control patient who maintains consistent propositional density throughout, attention should distribute relatively uniformly; for a Dementia patient whose language deteriorates mid-transcript, attention should concentrate on the degraded windows where fillers, incomplete propositions, and pronominal vagueness dominate.

This architecture offers RoBERTa a specific advantage over alternative long-context solutions. Rather than retraining or adapting the encoder itself to handle longer sequences (which risks destabilizing pre-trained representations), the sliding window approach treats RoBERTa as a fixed, high-quality local feature extractor and delegates the global aggregation to a lightweight recurrent module. This modular design means that RoBERTa's 160 GB worth of pre-trained linguistic knowledge [30] is fully preserved within each window, while the GRU, with only $\sim$2.6M parameters for a 2-layer bidirectional configuration with hidden dimension 256, learns solely the cross-window aggregation function. The parameter efficiency of this approach is critical in our data-scarce setting: instead of adapting 125 million encoder parameters to handle longer contexts, we train fewer than 3 million aggregation parameters on 234 training transcripts, dramatically reducing the overfitting risk.

The architecture also introduces a qualitatively new axis of interpretability absent from all other models in this study. Standard transformer classifiers produce a single prediction from a single [CLS] vector, offering no intrinsic mechanism to identify *which portions of the input* drove the classification. While post-hoc methods like SHAP can attribute importance to individual tokens, they cannot reveal the model's assessment of transcript-level structure, whether the diagnostic signal came from the opening description, a mid-interview tangent, or a late-transcript breakdown. Our architecture's per-window attention weights provide exactly this structural interpretability: they constitute a learned, human-readable "importance map" over the temporal progression of

the transcript. When combined with per-token SHAP attributions computed within each window, the result is a two-level diagnostic explanation: (1) *which transcript segments* the model considers most diagnostic (window attention), and (2) *which specific tokens within those segments* drive the prediction (SHAP). This hierarchical interpretability directly aligns with clinical practice, where speech-language pathologists evaluate both the global trajectory of a patient's discourse and the specific lexical and syntactic markers within individual utterances.

We trained this architecture in two configurations:

- *Frozen RoBERTa + GRU*: RoBERTa weights are frozen and only the GRU, attention, and classifier parameters are trained with Adam (`lr=1e-3`) and weighted cross-entropy loss over 50 epochs with batch size 16. This configuration isolates the contribution of the hierarchical aggregation from any encoder adaptation, testing whether RoBERTa's pre-trained [CLS] embeddings are sufficiently expressive for window-level classification when given proper cross-window contextualization.
- *End-to-End Fine-Tuned*: All parameters (RoBERTa encoder, GRU, attention, classifier) are jointly optimized with AdamW (`lr=2e-5`, `weight_decay=0.01`) and weighted cross-entropy loss over 10 epochs with batch size 4 (smaller due to backpropagation through multiple windows per sample). This configuration allows RoBERTa to adapt its per-window representations to be maximally informative for the downstream GRU aggregation, potentially learning to emphasize dementia-specific features in its [CLS] output.

6. **LLaMA Fine-Tuning**: LLaMA 3.1 is a popular open-weight decoding LLM for personal offline tasks, especially utilized among researchers. [35] By replacing the decoder model's causal attention mechanism with bi-directional attention, performing extensive fine-tuning using contrastive learning, and using global average pooling to generate fixed-size vector representations, Babakhin et al. enabled an instruction-aware, long context, multilingual sentence-level embedding LLM. [36]

   We integrated the state-of-the-art 8-Billion parameter LLM `nvidia/llama-embed-nemotron-8b` [36] model through two distinct pathways:

- *Frozen Base + MLP*: We extracted static 4,096-dimensional embeddings from the last token position of the final hidden layer (following the decoder-style convention for this architecture) and classified them through a PyTorch MLP with hidden layers `(64, 32)`, ReLU activations, and Dropout (`p=0.1`), trained with Adam (`lr=1e-3`) and weighted binary cross-entropy loss over 15 epochs with batch size 32.

- *Low-Rank Adaptation (LoRA)*: We instantiated LoRA [37] fine-tuning specifically targeting the `q_proj`, `k_proj`, `v_proj`, and `o_proj` attention modules with a rank of $r = 8$, $\alpha = 16$, and `lora_dropout=0.1`. A custom `NemotronForClassification` wrapper was implemented: the LoRA-adapted base model's final hidden states are mean-pooled across the sequence dimension (weighted by the attention mask), and the resulting 4,096-dimensional vector is projected through an `nn.Linear(4096, 2)` classification head. The model was trained end-to-end with AdamW (`lr=2e-5`, `weight_decay=0.01`) and weighted binary cross-entropy loss over 15 epochs with batch size 8, operating in `bfloat16` precision. LoRA is a Parameter Efficient Fine-Tuning (PEFT) technique that, instead of updating each attention block's full weight matrices, trains two lower-rank decomposition matrices ($A \in \mathbb{R}^{d \times r}$, $B \in \mathbb{R}^{r \times d}$) whose product is added to the frozen pre-trained weights. Since the fine-tuning operates only on these low-rank projections, LoRA enables comparable fine-tuning performance while substantially reducing the number of trainable parameters and alleviating computational constraints.

4.1.4 Interpretability Strategy

To explain and validate the predictions of the models, we employed the following interpretability techniques for different models trained:

- **Native Feature Importance**: Gain-based feature importance was extracted directly from the trained XGBoost gradient boosted tree models to identify the most discriminative TF-IDF unigrams and trigrams.

- **Permutation Importance [38]**: Used for the high-dimensional TF-IDF MLP models

to extract top predictive unigrams and trigrams by measuring the mean decrease in F1 score when each feature is randomly shuffled.

- **Integrated Gradients [25]**: Deployed on sequence models (CNN, BiLSTM) to trace gradient flows from the embedding layer to the output and attribute each token's contribution to the prediction.

- **SHAP (SHapley Additive exPlanations) [39]**: Utilized for BERT and Llama-Nemotron architectures to assign marginal contribution values to sub-word tokens by systematically masking each word in a sequence and evaluating its impact on the prediction.

- **Window Attention Visualization**: For the RoBERTa + GRU sliding window architecture, we visualized the learned per-window attention weights alongside the top SHAP-attributed tokens within each window. This dual-level interpretability reveals both *which transcript segments* the model prioritizes (via window attention) and *which tokens within those segments* drive the prediction (via SHAP), providing clinicians with a hierarchical view of the model's diagnostic reasoning.

Making models interpretable ensures a robust ML pipeline for this study, especially when we are dealing with such a scarce amount of data points in the DementiaBank Pitt Corpus. The interpretations of these models help us understand what features the models are attending to, which tells us if the model is finding spurious correlations within our small dataset or is predicting based on actual linguistic attributes.

4.2 Results

The deep learning experiments yielded detailed insights into both the performance and the learning behavior of cognitive decline predictors.

4.2.1 Model Performance Summary

Table 4.1 and Table 4.2 summarize the test-set performance of all models evaluated in this study. Three broad trends emerge from the results. First, rule-based propositional idea density (Idea Density LR) performs near chance (macro F1 = 0.493), confirming that raw

propositional counts are insufficient for dementia classification in transcribed speech. Second, TF-IDF and learned-embedding baselines (MLP, 1D CNN) achieve suspiciously perfect scores (F1 = 1.000), which, as we investigate in Section 4.2.3, reflects Clever Hans memorization of task-specific vocabulary rather than genuine linguistic understanding. Third, among the transformer-based models in Table 4.2, fine-tuned variants consistently outperform their frozen encoder counterparts across all four architectures, with RoBERTa Full Fine-Tuning and the two RoBERTa + GRU sliding window configurations emerging as the top-performing models (macro F1 $\geq$ 0.980, accuracy = 0.983). The subsequent subsections analyze each of these findings in detail.

Table 4.1: Performance of Various Non-LLM ANNs on predicting Dementia Transcripts in the Pitt Corpus

Model	**Acc.**	**Control**			**Dementia**			**Macro Average**		
		P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
Idea Density LR	0.508	0.344	0.500	0.408	0.666	0.512	0.579	0.505	0.506	0.493
XGBoost Unigrams	0.949	0.904	0.950	0.926	0.973	0.948	0.961	0.939	0.949	0.943
XGBoost Trigrams	0.898	0.888	0.800	0.842	0.902	0.948	0.925	0.895	0.874	0.883
MLP Unigrams	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
MLP Trigrams	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
1D CNN	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**
BiLSTM Attention	0.932	0.900	0.900	0.900	0.948	0.948	0.948	0.924	0.924	0.924

4.2.2 Idea Density Correlation

We conducted a comprehensive statistical evaluation of three established propositional idea density (PID) metrics: CPIDR, DEPID, and DEPID-R, to assess their discriminative capacity between the Dementia and Control cohorts. Point-biserial correlation coefficients and Welch's independent-samples t-tests were computed for each metric against the binary Dementia label.

Table 4.2: Performance of Various Transformer-based Models on predicting Dementia Transcripts in the Pitt Corpus

Model	**Acc.**	**Control**			**Dementia**			**Macro Average**		
		P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
DistilBERT Frozen + MLP	0.915	0.826	0.950	0.883	0.972	0.897	0.933	0.899	0.923	0.908
DistilBERT Full FT	0.949	0.869	1.000	0.930	1.000	0.923	0.960	0.934	0.961	0.945
RoBERTa Frozen + MLP	0.966	0.909	1.000	0.952	1.000	0.948	0.973	0.954	0.974	0.963
RoBERTa Full FT	**0.983**	**0.952**	**1.000**	**0.975**	**1.000**	**0.974**	**0.987**	**0.976**	**0.987**	**0.981**
ModernBERT Frozen + MLP	0.881	0.760	0.950	0.844	0.970	0.846	0.904	0.865	0.898	0.874
ModernBERT LoRA	0.932	0.833	1.000	0.909	1.000	0.897	0.945	0.916	0.948	0.927
RoBERTa Frozen + GRU	**0.983**	**0.952**	**1.000**	**0.975**	**1.000**	**0.974**	**0.987**	**0.976**	**0.987**	**0.981**
RoBERTa FT + GRU	**0.983**	**1.000**	**0.950**	**0.974**	**0.975**	**1.000**	**0.987**	**0.987**	**0.975**	**0.980**
Llama-Embed Frozen + MLP	0.881	0.933	0.700	0.800	0.863	0.974	0.915	0.898	0.837	0.857
Llama-Embed LoRA	0.915	0.857	0.900	0.878	0.947	0.923	0.935	0.902	0.911	0.906

Table 4.3: Statistical analysis of Propositional Idea Density metrics against the Dementia label.

Metric	r_{pb}	p **(corr.)**	t	p **(t-test)**
CPIDR	-0.1012	0.0838	-1.8447	0.0664
DEPID	$+0.0813$	0.1650	$+1.4926$	0.1369
DEPID-R	-0.0586	0.3172	-1.0027	0.3172

As reported in Table 4.3, none of the three PID metrics achieved statistical significance at the conventional $\alpha = 0.05$ threshold. CPIDR exhibited the strongest yet still non-significant negative correlation ($r_{pb} = -0.1012$, $p = 0.084$), with the corresponding t-test approaching marginal significance ($t = -1.845$, $p = 0.066$). This weak negative trend suggests that Dementia transcripts may exhibit slightly lower propositional density as measured by CPIDR, but the effect size is negligible. DEPID yielded a modest positive correlation ($r_{pb} = 0.081$, $p = 0.165$), while DEPID-R, which penalizes propositional repetitions, produced the weakest association ($r_{pb} = -0.059$, $p = 0.317$). The failure of DEPID-R to improve discriminability over DEPID indicates that repetition frequency alone does not reliably differentiate the two cohorts in transcribed speech.

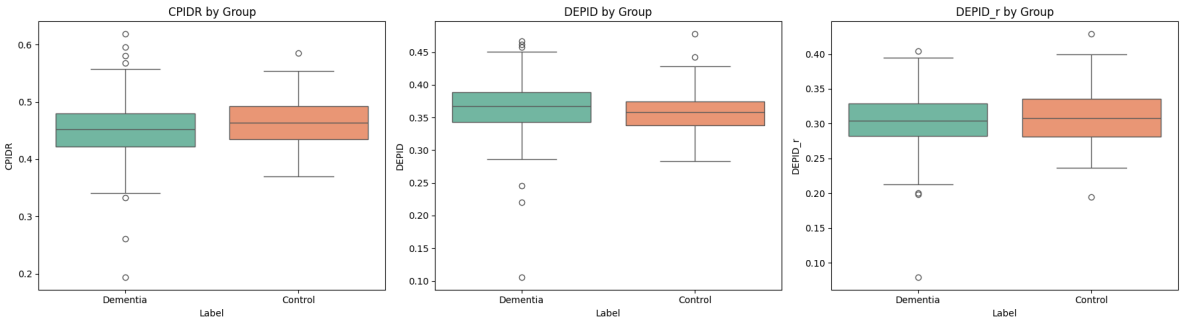

Figure 4.3: Box plots comparing the distribution of CPIDR, DEPID, and DEPID-R metrics between the Dementia and Control groups.

The box plots do not show that the prediction of Dementia vs Control group is associated with either of the 3 PID metrics, as the bodies of the boxes overlap on the y-axis.

Collectively, these results demonstrate that rule-based propositional idea density metrics, whether computed via shallow syntactic parsing (CPIDR), dependency structures (DEPID), or repetition-aware dependency parsing (DEPID-R), are insufficiently sensitive to the nuanced linguistic degradation patterns present in conversational speech transcripts. This underscores the necessity for learned, deep semantic representations capable of capturing higher-order pragmatic and syntactic deterioration beyond raw propositional counts.

4.2.3 Clever Hans Effect in High-Performing Baselines

The Clever Hans Effect [3] describes the phenomenon in which a model achieves higher performance by utilizing spurious correlations and artifacts present in the datasets rather than relying on causal relationships or task-related features.

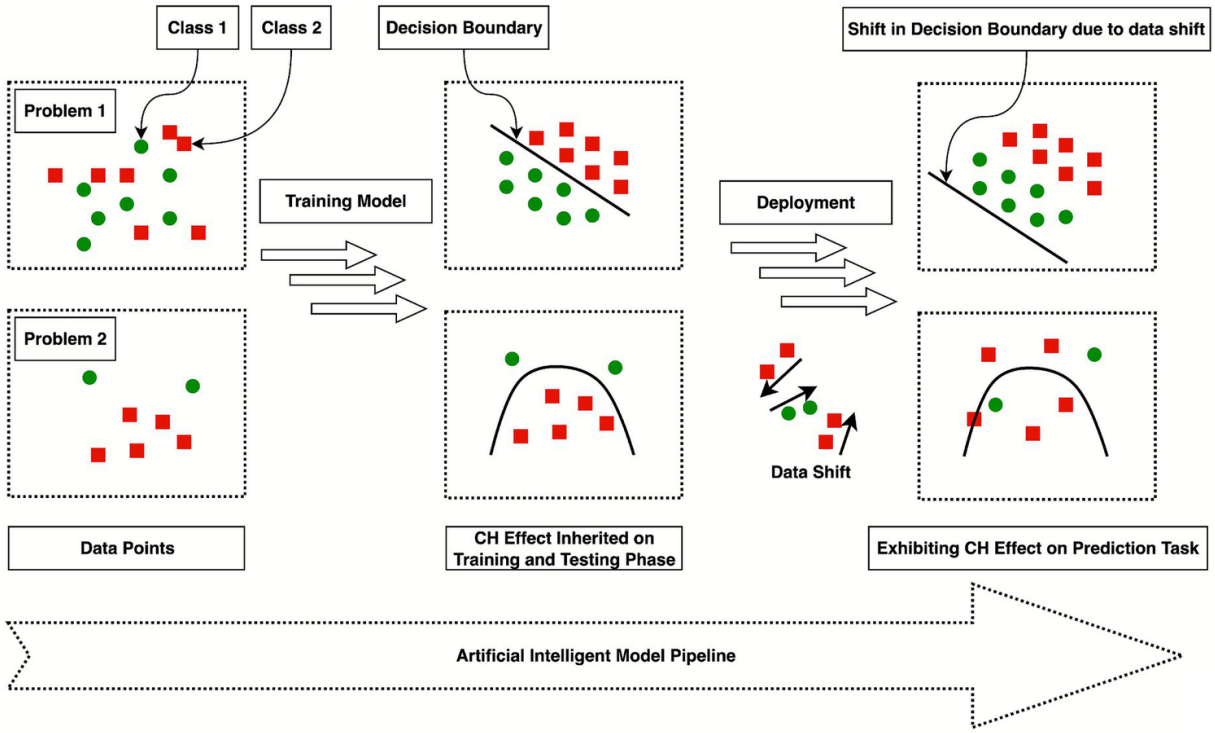

Figure 4.4: A classical example of the Clever Hans effect. [3]

As seen in Table 4.1, both the MLP and XGBoost TF-IDF baselines achieved anomalously high test scores (MLP: F1 = 1.000; XGBoost: F1 = 0.943), warranting rigorous investigation into the learned decision boundaries. We systematically examined the top predictive features across six high-performing baselines: XGBoost Unigram, XGBoost Trigram, MLP Unigram, MLP Trigram, 1D CNN, and BiLSTM with Attention, using native feature importance for tree-based models, Permutation Importance [38] for TF-IDF MLPs, and Integrated Gradients [25] for sequence models.

Below are the top 20 unigram and trigram features for XGBoost and MLP models as well as the Integrated Gradients attribution of 1D CNN and BiLSTM for model interpretation.

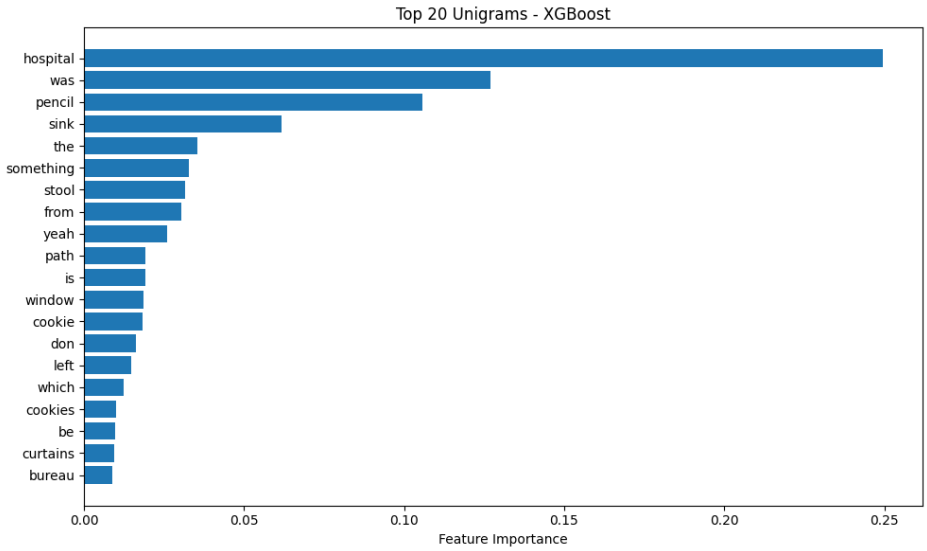

Figure 4.5: XGBoost Unigram: Top-20 feature importances. The dominant features are task-specific nouns (`hospital`, `pencil`, `sink`, `stool`, `cookie`) directly referencing the Cookie Theft stimulus.

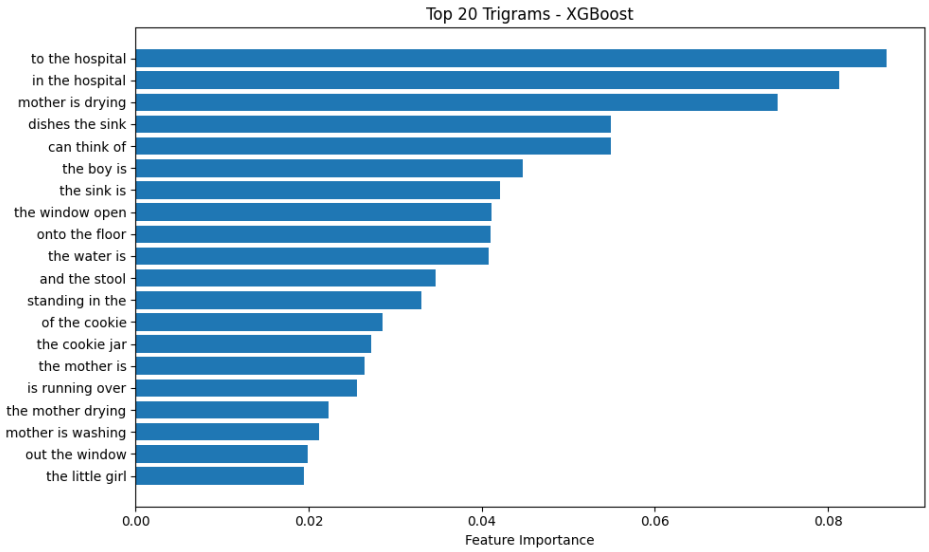

Figure 4.6: XGBoost Trigram: Top-20 feature importances. The highest-weighted trigrams (`to the hospital`, `in the hospital`, `mother is drying`) are stimulus-specific conversational tangents unique to the Dementia cohort's interview responses.

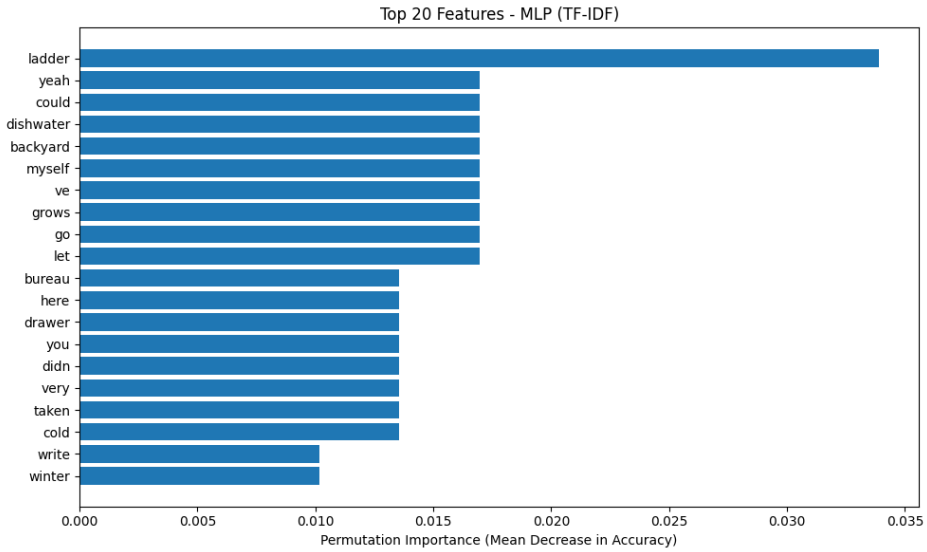

Figure 4.7: MLP Unigram: Top-20 permutation importance scores. The model's most predictive unigrams (`ladder`, `yeah`, `could`, `dishwater`) are task-specific vocabulary items rather than generalizable linguistic markers.

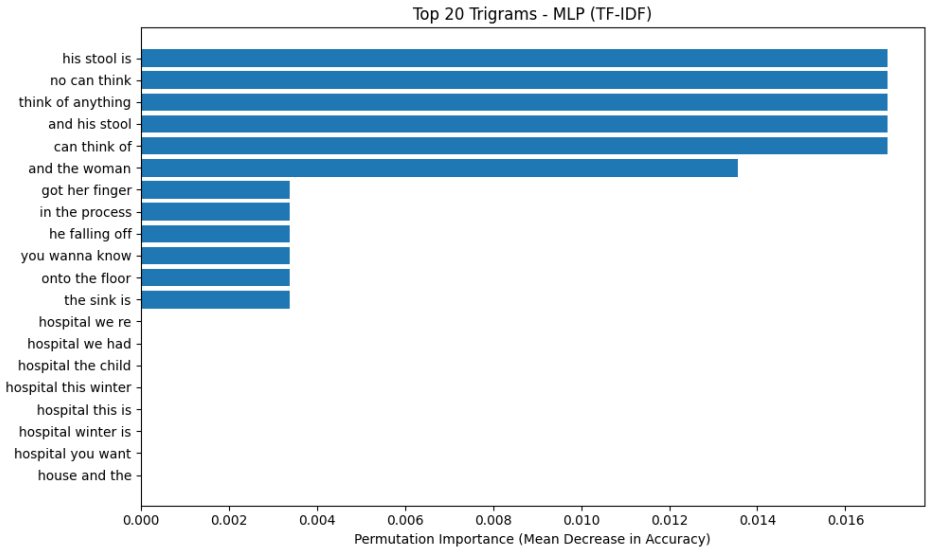

Figure 4.8: MLP Trigram: Top-20 permutation importance scores. Phrases such as `his stool is`, `no can think`, and `think of anything` confirm rote memorization of cohort-specific conversational tangents.

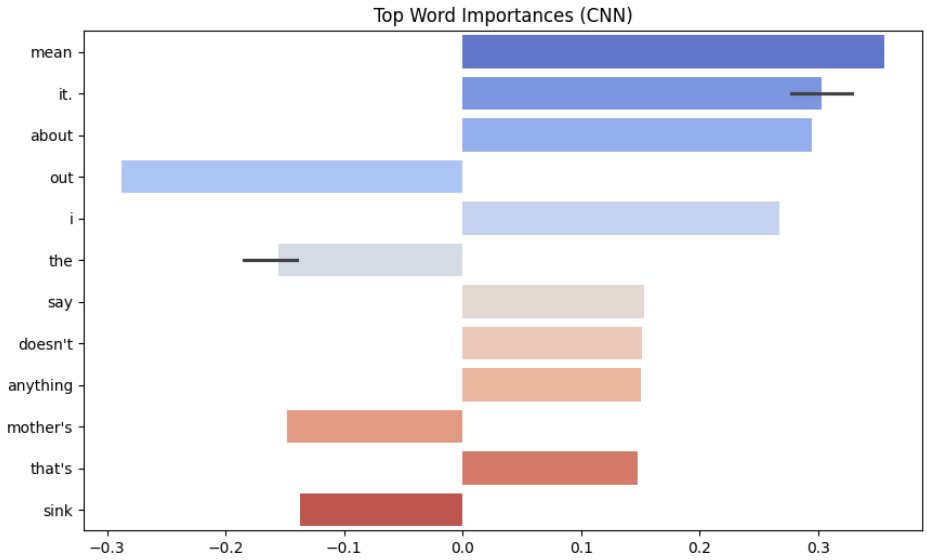

Figure 4.9: 1D CNN: Integrated Gradients attribution on a Dementia transcript. Positive attributions concentrate on vague, low-information tokens (`mean`, `it`, `about`, `i`, `say`), while negative attributions correspond to descriptive nouns (`mother's`, `sink`), suggesting the CNN distinguishes classes by the *absence* of propositional content.

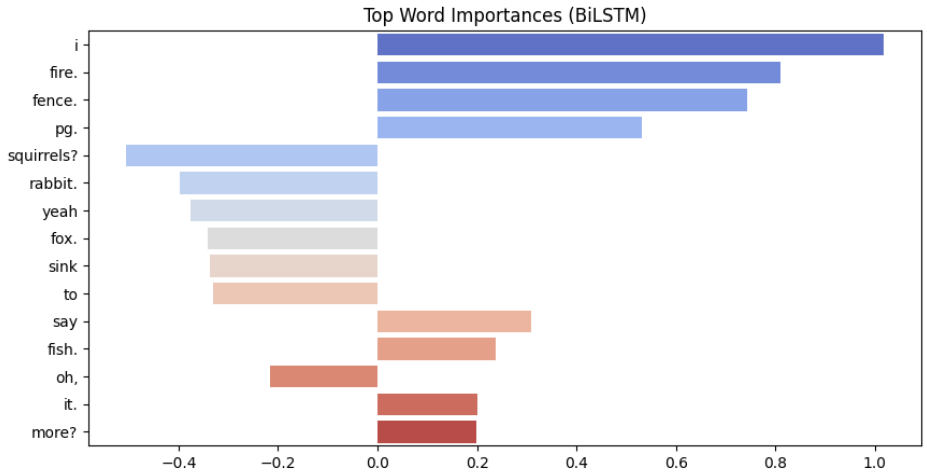

Figure 4.10: BiLSTM with Attention: Integrated Gradients attribution on a Dementia transcript. The highest attributions (`i`, `fire`, `fence`, `pg`) are idiosyncratic tokens from specific interview tangents, while negative attributions (`squirrels`, `rabbit`, `sink`) correspond to stimulus-descriptive vocabulary.

Across all six models, a consistent pattern emerges: the dominant predictive features are either (1) task-specific nouns and phrases directly referencing the Cookie Theft stimulus (`hospital`, `sink`, `cookie`, `mother is drying`), or (2) idiosyncratic conversational tangents specific to individual Dementia cohort interviews (`hospital this winter`, `fire`, `fence`). Crucially, the 1D CNN and BiLSTM with Attention, despite achieving 1.000 and 0.932 test accuracy, exhibit the same reliance on task-specific lexical cues as the MLP and XGBoost baselines, as evidenced by their Integrated Gradients attributions. This confirms that all non-LLM models operating on TF-IDF or learned-embedding token vocabularies are exploiting spurious correlations inherent in the Cookie Theft elicitation protocol rather than acquiring generalized representations of linguistic cognitive decline.

4.2.4 Deep Semantic and Pragmatic Model Performance

The pre-trained transformer and foundation models mitigated the "Clever Hans" effect by operating on sub-word tokenizations and contextualized embedding spaces that are decoupled from the task-specific TF-IDF vocabulary. As shown in Table 4.2, a consistent empirical pattern emerged across all encoder architectures: *fine-tuned models outperformed their frozen encoder + MLP counterparts*, demonstrating that domain adaptation through supervised fine-tuning (SFT) is essential for maximizing predictive performance even in data-scarce clinical settings.

- **DistilBERT**: Frozen + MLP (F1 = 0.908) → Full Fine-Tuning (F1 = 0.945), ΔF1 = +0.037.
- **RoBERTa**: Frozen + MLP (F1 = 0.963) → Full Fine-Tuning (F1 = 0.981), ΔF1 = +0.018.
- **ModernBERT**: Frozen + MLP (F1 = 0.874) → LoRA (F1 = 0.927), ΔF1 = +0.053.
- **Llama-Embed-Nemotron-8B**: Frozen + MLP (F1 = 0.857) → LoRA (F1 = 0.906), ΔF1 = +0.049.

This pattern confirms that while pre-trained encoders provide strong general-purpose representations, task-specific fine-tuning allows the model to adapt its internal representations to the nuanced linguistic markers of cognitive decline present in the Pitt Corpus. The

fine-tuning gains are most pronounced for smaller or less-adapted encoders (ModernBERT ΔF1 = +0.053, Llama-Nemotron ΔF1 = +0.049), while the already well-adapted RoBERTa shows a smaller but consistent improvement (ΔF1 = +0.018).

RoBERTa emerges as the optimal encoder architecture for this data regime. Among standard encoder configurations, **RoBERTa Full Fine-Tuning** achieved the highest performance (macro F1 = 0.981, accuracy = 0.983), substantially outperforming both the larger Llama-Embed-Nemotron-8B (LoRA F1 = 0.906) and the longer-context ModernBERT (LoRA F1 = 0.927). This advantage is attributable to RoBERTa's favorable position in the trade-off between representational capacity and adaptability: its 125M parameters provide sufficient expressiveness to capture the semantic and pragmatic features of cognitive decline, while remaining small enough to be fully fine-tuned on 234 training samples without catastrophic overfitting. In contrast, the 8-billion parameter Llama-Embed-Nemotron-8B, despite operating under LoRA's parameter-efficient constraints, introduces millions of trainable parameters that exceed the effective information content of the training set, leading to suboptimal generalization. ModernBERT's 8,192-token context window, while theoretically advantageous for long transcripts, did not translate into superior performance, suggesting that the critical diagnostic information in Pitt Corpus transcripts is concentrated in localized linguistic patterns rather than long-range dependencies.

The RoBERTa + GRU sliding window architectures matched the best performance. Both the frozen and end-to-end fine-tuned RoBERTa + GRU configurations achieved macro F1 = 0.981 and 0.980 respectively, tying with RoBERTa Full Fine-Tuning as the top-performing models. Notably, the RoBERTa Frozen + GRU variant achieved this without updating any RoBERTa encoder weights, only the GRU, attention, and classifier parameters were trained. The end-to-end variant exhibited a complementary strength profile: perfect Dementia recall (1.000) with slightly lower Control recall (0.950), compared to the frozen variant's perfect Control recall (1.000) with slightly lower Dementia recall (0.974). The success of the sliding window approach validates the hypothesis that hierarchical, window-level processing with learned attention aggregation is an effective strategy for handling long clinical transcripts. The per-window attention weights provide an additional interpretability mechanism, directly revealing which transcript segments the model considers

most diagnostic, a capability absent from standard fine-tuned encoders that truncate input to 512 tokens.

4.2.5 Case Study: Fine-Tuning Amplifies Diagnostic Signal

To understand how fine-tuning transforms a pre-trained encoder's internal decision-making, we applied SHAP (SHapley Additive exPlanations) to both the RoBERTa Frozen + MLP and RoBERTa Full Fine-Tuning models on the *same two transcripts*: a Dementia-labeled sample and a Control-labeled sample (ID 114). SHAP decomposes the final prediction into the additive contributions of each individual sub-word token, effectively highlighting which words and phrases drive the classification towards Control or Dementia. By comparing the SHAP attribution landscapes of the frozen and fine-tuned models on identical inputs, we can directly observe how supervised fine-tuning reshapes the encoder's attention to linguistic features.

Figure 4.11: SHAP attributions from the **RoBERTa Frozen + MLP** model on a Dementia transcript (top) and Control transcript ID 114 (bottom). Despite correct predictions, the SHAP values are near-zero across virtually all tokens, indicating a flat, undifferentiated attribution landscape where no individual token contributes meaningfully to the prediction.

Fig. 4.11 visualizes the token-level SHAP attributions from the frozen RoBERTa encoder. Two features are immediately apparent. First, the SHAP magnitudes are vanishingly small: nearly every token receives an attribution of ≈ 0.0, producing a flat, featureless attribution landscape. Although the frozen model achieves a correct Dementia prediction (P(Dem) = 0.993) and a correct Control prediction for ID 114 (P(Ctrl) = 0.875), it does so by relying on the diffuse geometry of the pre-trained embedding space rather than on sharply discriminative token-level features. The model cannot clearly articulate *which* tokens drove its decision. Second, the sparse tokens that do receive non-trivial attributions in the Dementia transcript

include dataset-specific vocabulary items, “rabbit”, “squirrels”, “fish”, “bear”, “wolves”, which are artifacts of the idiosyncratic conversational tangents in the Pitt Corpus interviews (where the interviewer prompts “what else do you see?” and Dementia patients fabricate animals not present in the Cookie Theft picture). These attributions mirror the Clever Hans pattern identified in the non-LLM baselines (Section 4.2.3), suggesting that even the frozen RoBERTa encoder partially relies on spurious, task-specific lexical correlations rather than on the deeper linguistic markers of cognitive decline.

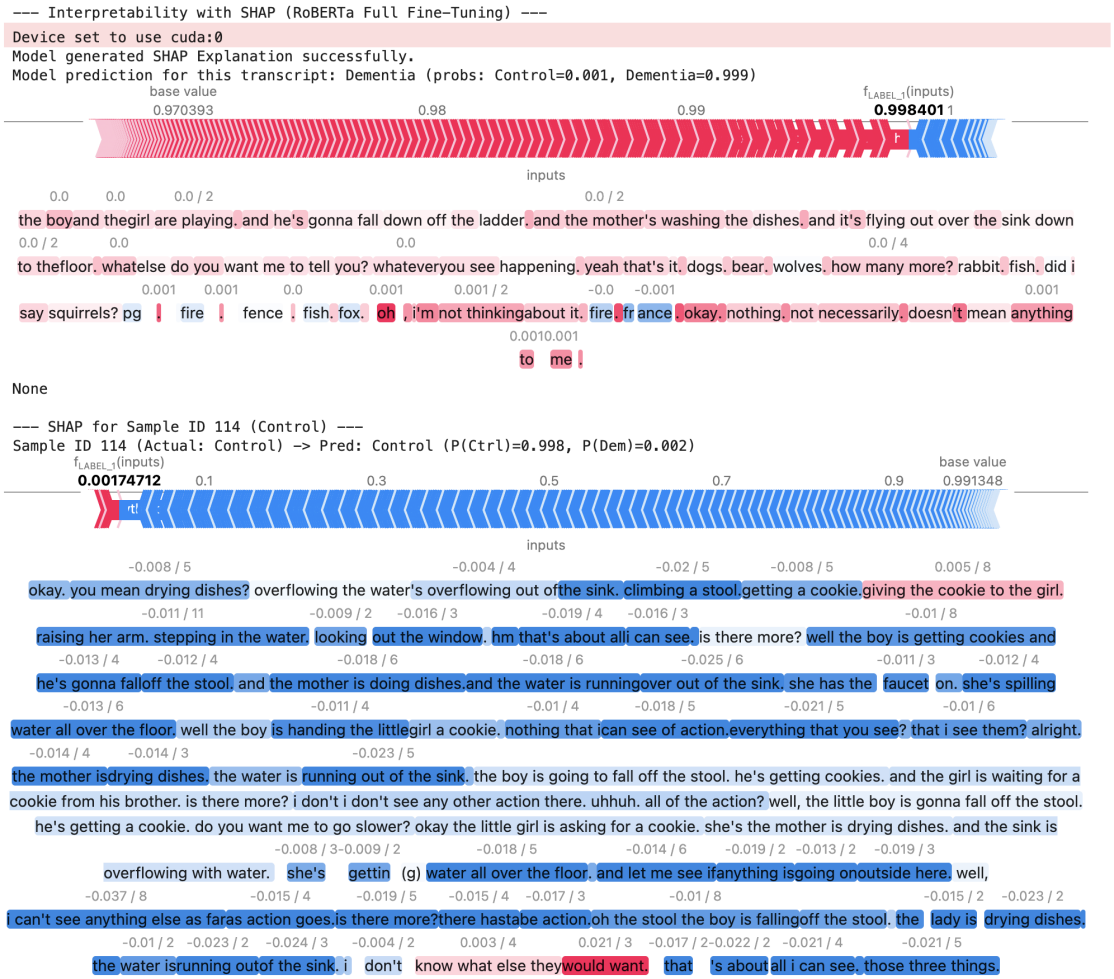

Figure 4.12: SHAP attributions from the **RoBERTa Full Fine-Tuning** model on the same Dementia transcript (top) and Control transcript ID 114 (bottom). Fine-tuning produces dramatically stronger, more differentiated attributions: dense blue (Control-pushing) signal on propositional content and strong red (Dementia-pushing) signal on hesitation markers and empty phrases.

Fig. 4.12 reveals a fundamentally different attribution landscape after fine-tuning. On the same Dementia transcript, the fine-tuned model (P(Dem) = 0.999) now produces strong, concentrated SHAP attributions on linguistically meaningful tokens. Critically, the dataset-specific animal vocabulary ("squirrels", "rabbit", "fish", "bear", "wolves") that dominated the frozen model's sparse attributions now receives near-zero SHAP values, fine-tuning has effectively *de-emphasized* these spurious lexical correlations. Instead, the model's Dementia-pushing (red) attributions concentrate on pragmatically diagnostic phrases: "oh", "i'm not thinking about it", "okay", "nothing", "not necessarily", "doesn't mean anything to me". These are precisely the markers of cognitive decline identified in the clinical literature: empty discourse fillers, explicit expressions of inability to perform the task, and semantically vacuous hedging that signals the patient's awareness of their own communicative failure without the capacity to produce substantive propositional content.

On the Control transcript (ID 114), the contrast is equally striking. Where the frozen model produced weak, diffuse blue attributions (P(Ctrl) = 0.875), the fine-tuned model generates dense, saturated blue signal across virtually every propositionally rich phrase (P(Ctrl) = 0.998): "you mean drying dishes?", "overflowing the water's overflowing out of the sink", "climbing a stool", "getting a cookie", "giving the cookie to the girl", "raising her arm", "stepping in the water", "looking out the window", "the mother is drying dishes", "the water is running out of the sink", "the boy is going to fall off the stool", "he's getting cookies", and "the girl is waiting for a cookie from his brother". Each of these phrases contains specific action verbs, concrete nouns, and complete syntactic structures are the hallmarks of high propositional idea density that characterize cognitively healthy speech. The fine-tuned model has learned to recognize these structural, idea-filled sentences as strong evidence against Dementia, assigning them large negative SHAP values that collectively push the prediction decisively toward Control.

This side-by-side comparison provides direct interpretive evidence for why fine-tuning outperforms frozen encoders in our results (Table 4.2). The frozen encoder operates on a general-purpose embedding space where the distinction between Dementia and Control transcripts is encoded as a subtle geometric separation which are sufficient for above-chance classification but producing weak, undifferentiated token attributions. Fine-tuning reshapes

this embedding space through task-specific gradient updates, amplifying the encoder's sensitivity to the syntactic and semantic markers that are genuinely diagnostic of cognitive decline while suppressing attention to dataset-specific vocabulary artifacts. The result is not only higher accuracy (F1: 0.963 $\rightarrow$ 0.981) but also more clinically interpretable predictions: a fine-tuned model whose SHAP attributions directly align with established linguistic theories of propositional idea density and discourse coherence.

4.2.6 Case Study: Window Attention in RoBERTa + GRU

The RoBERTa Frozen + GRU and RoBERTa FT + GRU architectures achieved near-identical macro F1 scores (0.981 vs. 0.980), yet this numerical parity conceals a fundamental qualitative difference in *how* the two models distribute attention across sliding windows. To investigate this, we visualized the learned per-window attention weights for five Dementia and five Control transcripts under both the frozen and end-to-end fine-tuned configurations, selecting the longest transcripts in each class to maximize the number of windows available for comparison.

4.2.6.1 Frozen RoBERTa + GRU: Recency Bias

As shown in Fig. 4.13 and Fig. 4.14, the frozen RoBERTa + GRU model exhibits a pronounced *recency bias*: the learned attention weights consistently concentrate on the final one or two windows of each transcript, regardless of whether the transcript is Dementia or Control, and regardless of where the most diagnostically informative content actually resides. This pattern is consistent across all ten visualized transcripts.

The recency bias admits a principled explanation rooted in the interaction between the frozen encoder and the GRU's sequential inductive bias. When RoBERTa's weights are frozen, all window-level [CLS] embeddings are drawn from the same generic pre-trained representation space. These embeddings are not optimized for the downstream classification task, so the semantic differences between windows containing diagnostically rich content (e.g., fragmented tangential speech) and windows containing neutral content (e.g., interviewer prompts) are subtle. The bidirectional GRU, tasked with aggregating these weakly differentiated embeddings, defaults to the simplest strategy that minimizes training loss: relying on the final hidden states, which encode a cumulative summary of the entire transcript

through the GRU's recurrent dynamics. This is analogous to how a standard unidirectional RNN classifier uses only its last hidden state, so the GRU's attention mechanism learns to approximate this behavior by assigning disproportionate weight to the terminal windows.

We hypothesize that the frozen model's strong overall performance (F1 = 0.981) is primarily attributable to the fact that the sliding window architecture now provides the classifier access to the *full transcript content* rather than merely the first 512 tokens. In effect, the frozen RoBERTa + GRU achieves its performance gain not through sophisticated window-level reasoning, but through a simpler mechanism: the GRU's final hidden state captures a holistic summary of all windows, including the late-transcript content that the standard 512-token truncation would have discarded entirely. The attention mechanism, rather than performing genuine selective attention over transcript regions, merely confirms our hypothesis by revealing that the recurrent pathway, not the attention pathway, is doing the heavy lifting.

4.2.6.2 Fine-Tuned RoBERTa + GRU: Content-Driven Attention

Fig. 4.15 and Fig. 4.16 reveal a qualitatively different attention pattern under end-to-end fine-tuning. The recency bias observed in the frozen configuration is substantially reduced or eliminated: attention weights are distributed more dynamically across the full span of the transcript, with peaks that vary per-sample rather than systematically clustering at the final windows. For Dementia transcripts, the fine-tuned model's attention concentrates on windows containing the highest density of hesitation markers, empty fillers, and fragmented syntactic structures are the same token-level features identified as diagnostic by the SHAP analysis in the preceding case study. For Control transcripts, attention shifts toward windows rich in propositional content: concrete descriptions of the Cookie Theft scene with specific action verbs and complete syntactic structures.

This content-driven attention distribution emerges because end-to-end fine-tuning allows the RoBERTa encoder to produce [CLS] embeddings that are differentiated with respect to the downstream task. When the encoder weights are updated jointly with the GRU, the per-window [CLS] vectors are no longer generic representations from a shared pre-trained space; instead, they become task-specialized embeddings where windows containing diagnos-

tic content produce representations that are geometrically distinct from windows containing neutral content. The additive attention mechanism can then meaningfully discriminate between windows, assigning higher weights to those whose fine-tuned [CLS] embeddings carry the strongest diagnostic signal.

The practical implication is significant: although the frozen and fine-tuned RoBERTa + GRU configurations achieve comparable aggregate performance (F1 = 0.981 vs. 0.980), they arrive at their predictions through fundamentally different mechanisms. The frozen model relies on a position-biased recurrent summary that happens to capture sufficient global information for accurate classification. The fine-tuned model implements genuine content-based selective attention over transcript regions, producing window-level importance weights that are both more dynamic across samples and more clinically interpretable. For deployment in clinical decision-support systems, where interpretability and robustness to varying transcript lengths and structures are essential, the fine-tuned RoBERTa + GRU configuration is therefore the preferred architecture despite the numerical parity in aggregate F1.

4.3 Conclusion

Our evaluation confirms the initial hypothesis: linguistic ability reflects neurological development, and modern pre-trained transformer architectures are vastly superior at extracting subtle markers of cognitive decline compared to traditional TF-IDF classifiers, sequential ANNs, or rule-based propositional idea density computations. Systematic comparisons across four encoder architectures (DistilBERT, RoBERTa, ModernBERT, and Llama-Embed-Nemotron-8B) consistently demonstrated that fine-tuning, whether full-parameter or parameter-efficient via LoRA, yields superior performance compared to frozen encoder baselines, confirming that domain adaptation through supervised fine-tuning is essential even on datasets of this scale. **RoBERTa Full Fine-Tuning** and the **RoBERTa + GRU sliding window architectures** emerged as the strongest configurations, all achieving macro F1 $\geq$ 0.980 and accuracy of 0.983. RoBERTa's 125M parameter count represents an optimal trade-off between representational capacity and adaptability for scarce clinical data: large enough to capture nuanced semantic features of cognitive decline, yet small enough for stable full fine-tuning on 234 training samples.

The RoBERTa + GRU sliding window approach is particularly noteworthy, as it addresses the fundamental information loss caused by the 512-token truncation limit of standard BERT-family encoders. By partitioning transcripts into overlapping windows and aggregating them through a GRU with learned attention, this architecture preserves the full transcript content while providing window-level interpretability that directly reveals which segments of a patient's speech the model considers most diagnostic. The comparable performance of the frozen and end-to-end variants (F1 = 0.981 vs. 0.980) suggests that RoBERTa's pre-trained representations are sufficiently expressive for this hierarchical aggregation strategy, and that similar sliding window techniques should be adopted as a standard approach for efficient long clinical text processing in predictive clinical NLP pipelines.

SHAP interpretability confirmed that these models attend to theoretically grounded linguistic markers: propositional density and specific descriptive vocabulary in Control transcripts, versus conversational fillers, vague pronominal references, and fragmented syntactic structures in Dementia transcripts. However, the performance ceiling and the reliance on speech-specific artifacts indicate that substantially larger and more diverse clinical corpora will be required for these models to generalize across elicitation protocols (e.g., picture description vs. autobiographical writing vs. life-narrative interviews) and detect cognitive decline regardless of the specific linguistic task.

4.4 Discussion and Future Work

The SHAP interpretability analysis reveals both the promise and the current limitation of LLM-based cognitive decline prediction. On one hand, the fine-tuned RoBERTa encoder demonstrably attends to linguistically principled features, propositional density, lexical specificity, and syntactic coherence, that align with established clinical markers of Alzheimer's disease. On the other hand, the model's concurrent reliance on speech-specific artifacts such as conversational fillers ("well", "ah", "uh") and hedging conjunctions ("and", "so") indicates that the learned representations remain tethered to the distributional characteristics of transcribed spoken language. This domain specificity constitutes the primary obstacle to deploying these models for written text analysis: the linguistic markers of cognitive decline in autobiographical writing, formal essays, or clinical narratives differ substantially from those

manifested in semi-structured interview transcripts.

The consistent advantage of fine-tuning over frozen encoders across all four architectures demonstrates that even with only 234 training transcripts, supervised domain adaptation successfully specializes pre-trained representations to the clinical task. However, the magnitude of the fine-tuning benefit is modulated by model size: RoBERTa (125M parameters) achieves the largest absolute performance under fine-tuning (F1 = 0.981), while the much larger Llama-Embed-Nemotron-8B (8B parameters) shows improvement with LoRA but reaches a lower ceiling (F1 = 0.906). This suggests that for data-scarce clinical NLP, moderately-sized encoders like RoBERTa occupy an optimal point in the bias-variance trade-off: they possess sufficient capacity to learn domain-specific features through fine-tuning without the overfitting risk inherent in adapting billion-parameter models to hundreds of training samples.

The success of the RoBERTa + GRU sliding window architecture deserves particular attention. By decomposing long transcripts into overlapping 512-token windows and aggregating them through a recurrent network with learned attention, the architecture simultaneously solves the information truncation problem and introduces a new axis of interpretability. The window-level attention weights directly reveal which segments of a transcript contribute most to the classification, enabling clinicians to identify specific conversational passages that exhibit diagnostic linguistic degradation. This hierarchical processing paradigm should be adopted more broadly in clinical NLP contexts where documents exceed standard transformer context limits, as it preserves complete textual information while remaining computationally tractable.

The Nun Study [6], which longitudinally tracked the relationship between early-life autobiographical writing quality and late-life Alzheimer's incidence, represents a particularly compelling data source for future work. Training on the Nun Study autobiographies alongside transcribed speech corpora such as DementiaBank would expose the model to both written and spoken registers of cognitively impaired language, potentially enabling the cross-modal generalization that the current Pitt Corpus alone cannot support. Additional longitudinal datasets spanning multiple elicitation protocols: structured clinical interviews, free-form narrative tasks, and written correspondence, would further diversify the training distribution.

Ultimately, this work serves as a preliminary investigation establishing (1) the inadequacy of rule-based propositional idea density metrics for transcribed speech, (2) the susceptibility of high-capacity non-LLM models to Clever Hans memorization on small clinical datasets, (3) the effectiveness of supervised fine-tuning for domain adaptation even on scarce clinical data, with RoBERTa's 125M-parameter architecture emerging as optimally sized for this data regime, (4) the viability of hierarchical sliding window architectures with attention-based aggregation for efficient long clinical text processing, and (5) the critical data bottleneck that must be resolved before these models can generalize across elicitation modalities. Future research should prioritize the aggregation of large-scale, multi-modal clinical language corpora and investigate curriculum learning strategies that progressively expose foundation models to increasingly diverse linguistic manifestations of cognitive decline.

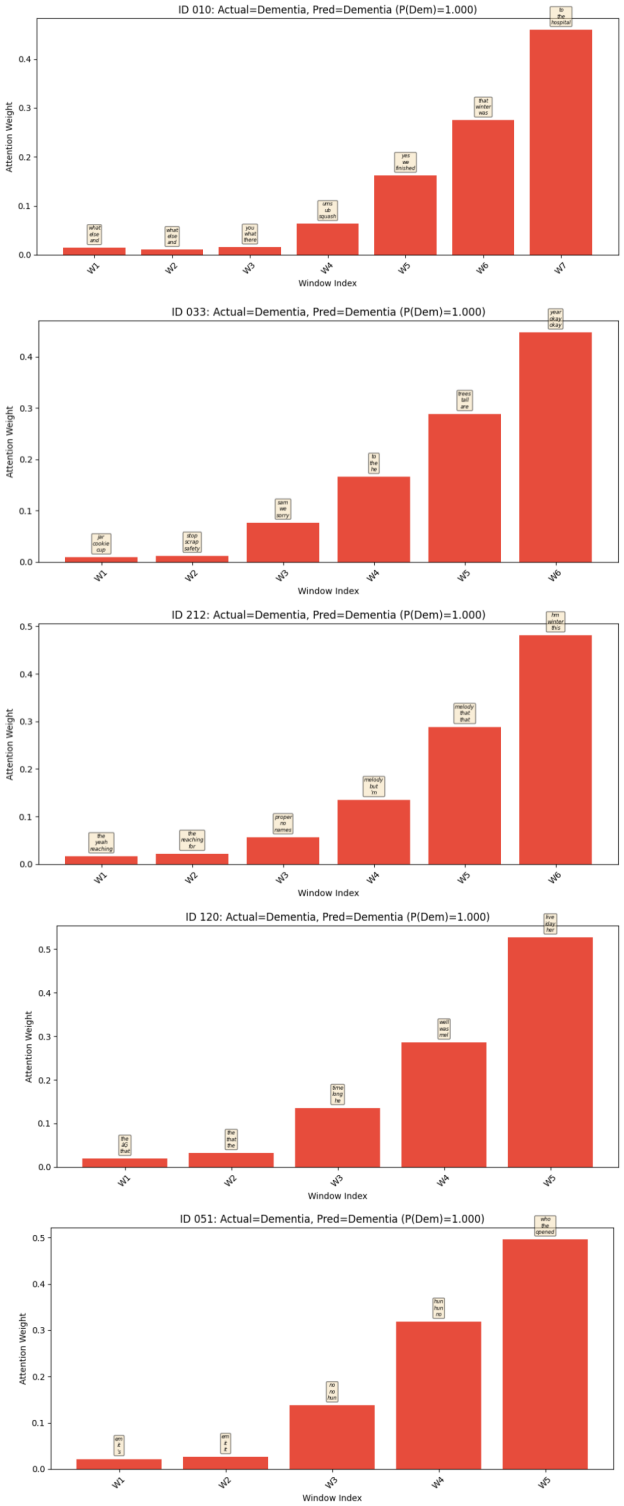

Figure 4.13: Window attention weights for 5 Dementia transcripts under **RoBERTa Frozen + GRU**. Attention concentrates heavily on the final windows of each transcript, exhibiting a strong recency bias regardless of where the most diagnostically relevant content appears.

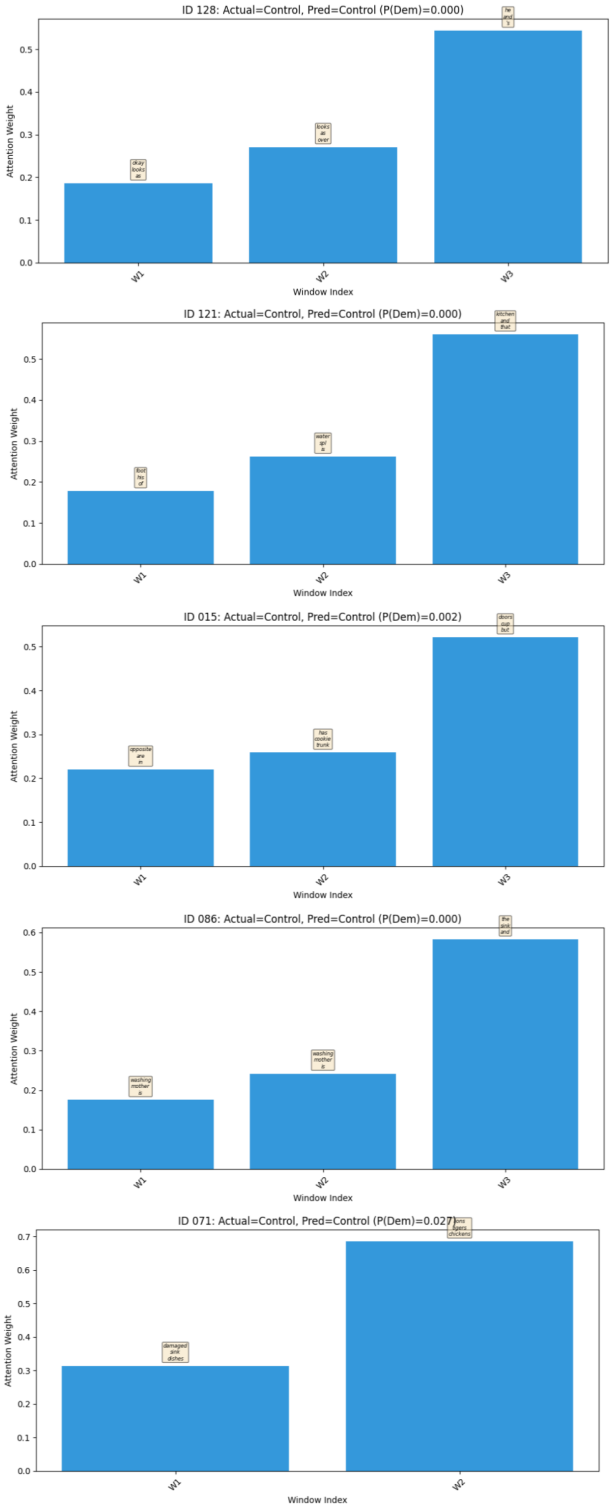

Figure 4.14: Window attention weights for 5 Control transcripts under **RoBERTa Frozen + GRU**. The same recency bias is observed: attention clusters on the final 1–2 windows irrespective of the propositional content distribution across the transcript.

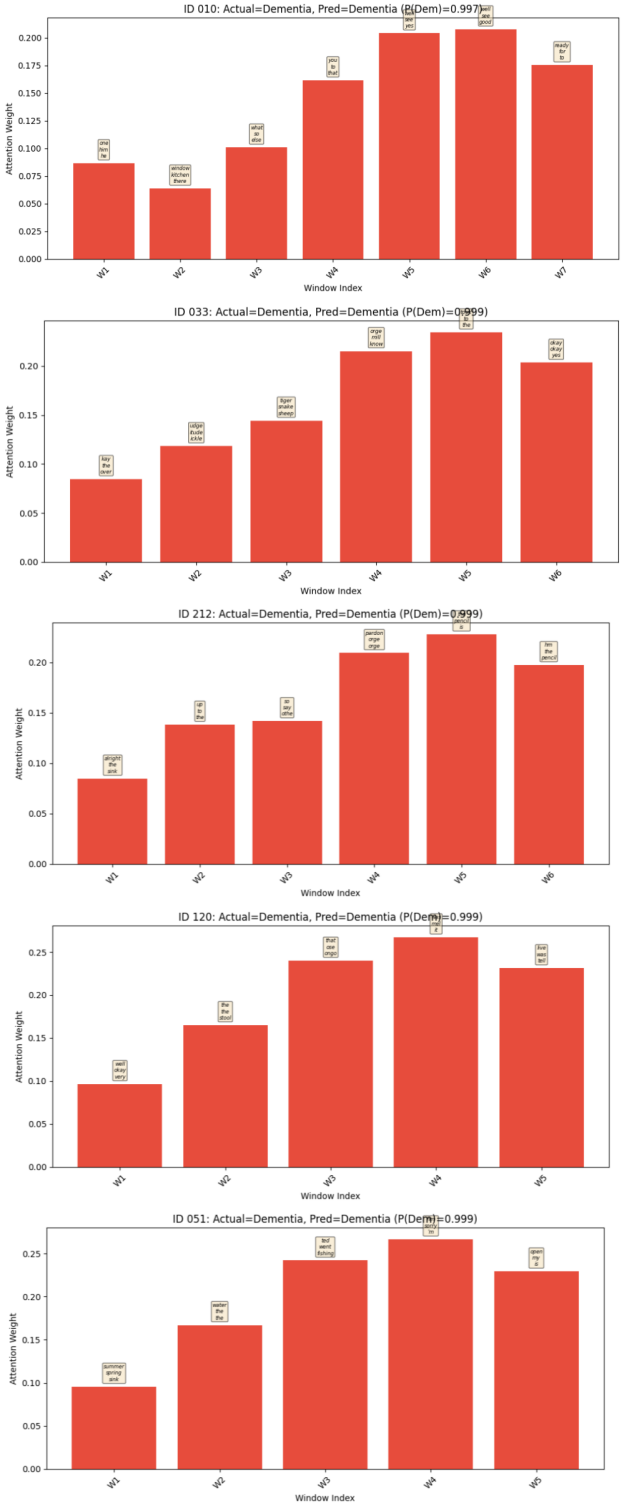

Figure 4.15: Window attention weights for 5 Dementia transcripts under **RoBERTa FT + GRU**. Attention is distributed more dynamically across windows, concentrating on segments that contain the most diagnostically relevant content rather than defaulting to the final windows.

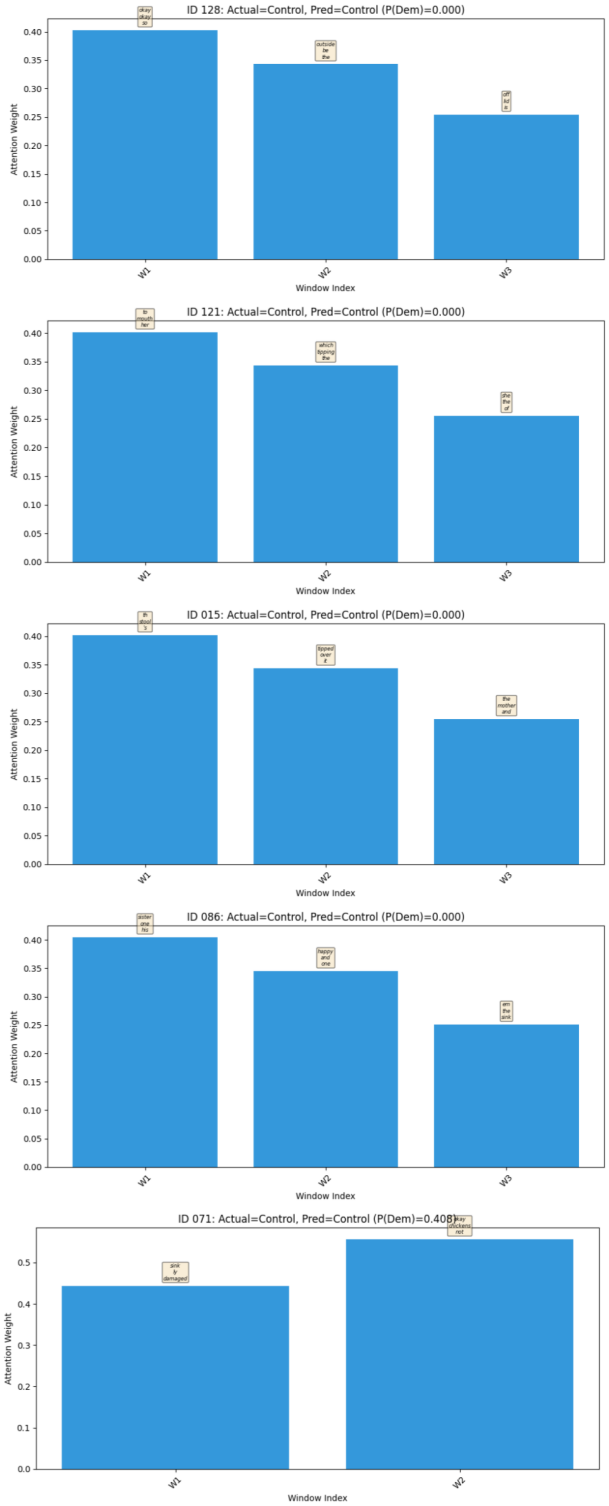

Figure 4.16: Window attention weights for 5 Control transcripts under **RoBERTa FT + GRU**. Attention distributes across windows containing propositionally dense descriptions, reflecting content-driven rather than position-driven selection.

Chapter 5

Longitudinal Cognitive Ability Trend

To evaluate whether human language idea density is declining at a population scale, we performed a longitudinal trend analysis across seven massive digital text archives. While our baseline analysis began with measuring general digital dialogues and web patterns via Reddit and Common Crawl, the scale of this experiment was vastly expanded. We incorporated Hacker News, academic articles from ICLR, life-science journals via BMC, and long-standing niche communities on Airliners.net and TheForce.net. This expansive approach guarantees that structural changes in human semantic communication can be mapped across radically different societal and technical ecosystems over extended periods.

5.1 Corpora Selection

Rather than relying on randomized sub-sampling, we processed the entirety of the available corpora to guarantee absolute statistical representation besides Common Crawl and Reddit. For every datapoint within each dataset's yearly distribution, we calculated the CPIDR (Computerized Propositional Idea Density Rater) [12] and DEPID [8] (Dependency-based Propositional Idea Density) scores. We then aggregated these metrics across all extracted data to determine the true mean and standard error of the mean (SEM) for each given year.

5.1.1 Reddit Comments

The Reddit corpus acts as the baseline for user-generated social conversational text. The dataset, spanning from 2007 to 2023, was systematically constructed by extracting standardized random samples of up to 500,000 text records per year from historical site archives. To ensure comprehensive longitudinal coverage, data collection pipelines were adapted to accommodate the storage formats of different temporal spans. Historical comments (2007–2011) were decompressed directly from .bz2 Pushshift project [40] hosted on

the Internet Archive [41], while intermediate records and submissions (2012–2018) were programmatically streamed from .parquet mirrors hosted on Hugging Face [42]. For the most recent era (2019–2023), raw JSON outputs were decompressed on-the-fly from large-scale .zst archives from Hugging Face [43]. Across all distinct data sources, a unified preprocessing schema was applied: text payloads were cleaned of extraneous whitespace and random ascii characters, explicitly filtered to remove deleted, removed, or automated bot inputs, and subjected to a 20-character minimum length threshold before being securely concatenated into plain-text yearly compilation files for subsequent analysis. We collected this to monitor trends in broad-spectrum social communication across varying, generally younger demographics mixed with computer and smartphone use of the website.

5.1.2 Common Crawl

To represent the broader evolution of general internet discourse, longitudinal web data spanning from 2013 to 2024 was systematically extracted from the Common Crawl repository [44]. For each target year, a representative Web ARChive (.warc.gz) segment from the flagship CC-MAIN crawl was streamed directly from the official AWS data repository. Data ingestion was managed programmatically using the warcio library, iteratively decompressing the stream to isolate HTTP response records. To ensure the linguistic quality of the corpus, the raw HTML payloads were subjected to a rigorous parsing pipeline: structural HTML tags and character entities were stripped, and excessive whitespace was normalized. The resulting text was filtered to exclude fragmented lines (fewer than 20 characters), and only substantial documents yielding over 100 characters of clean text were retained. This extraction protocol was capped at a standardized sample size of 5,000 valid documents per year, culminating in discrete, yearly plain-text corpora designed for longitudinal density analysis. The Common Crawl dataset captures the overall landscape of the internet, excluding highly specific silos, providing an "average" web density baseline. For this reason the Common Crawl is commonly used to pre-train modern LLMs. Sourcing general web pages, unstructured blogs, and assorted domains provided a macro-level view of standard internet exposition.

5.1.3 Airliners.net all fora comments

To construct an ecologically valid dataset of domain-specific discourse, textual data was retrieved from the digital aviation community forum, Airliners.net [45]. Since no programmatic API or centralized data repository exists for this platform, a custom, high-throughput asynchronous web scraping pipeline was engineered. The extraction architecture utilized a dynamic thread pooling methodology over a robust SQLite persistence layer, guaranteeing operational idempotency across the expansive, nested pagination hierarchies of the site's subforums and comment threads. Raw HTML payloads were incrementally fetched and computationally parsed using the BeautifulSoup library [46]. A stringent data extraction heuristic was employed to systematically strip away nested quoting elements (<blockquote>), ensuring that only original textual contributions were preserved. Additionally, cascading regular expression fallbacks were instituted to accurately capture the chronological timestamp of each post while bypassing structural aberrations in the forum's Document Object Model (DOM). The resulting cleaned comments and associated metadata were securely segregated by extraction year into discrete .jsonl (JSON Lines) files. The text collected heavily consists of aviation enthusiast and airline industry professional conversations. This domain was targeted because it represents a long-standing, niche hobbyist community boasting deep domain knowledge and exceptionally low user-churn dating back to 1998.

5.1.4 Hacker News all fora comments

To capture the specialized discourse of the technology and startup sectors, longitudinal comment data spanning from 2005 to the present (2026) was extracted from the social news aggregation platform, Hacker News [47]. Rather than relying on rate-limited web scraping or native APIs, data ingestion was executed using the official Hacker News datasets hosted within the Google Cloud Public Datasets program [48]. A programmatic extraction pipeline was implemented in Python using the google-cloud-bigquery client architecture to authenticate and query the bigquery-public-data.hacker_news.full table. Optimized SQL queries were generated dynamically to partition the data by year, sequentially filtering the database for records specifically classified as computationally significant textual comments (type = 'comment'). Along with the payload text, pertinent structural metadata including

the item identifier, author pseudonym, community score, and parent-child hierarchical relations were preserved. The fetched records were structured into chunked pandas DataFrames and exported to standard comma-separated value (.csv) formats, which were subsequently processed into homogeneous JSON Lines (.jsonl) archives to support the downstream longitudinal density analysis. As a technology, startup, and computer science-focused forum, this dataset captures a highly moderated and deeply technical community demographic famously known for long-form discussion.

5.1.5 TheForce.net all fora comments

To obtain a robust, culturally specific linguistic corpus, textual data was sourced from the protracted, domain-specific discussions on the fan-run forums at TheForce.net [49]. In the absence of a native programming interface—I find their lack of an API disturbing—a high-throughput, asynchronous web scraping architecture was developed to systematically traverse and ingest the site's hierarchical sub-forums and thread paginations. The extraction pipeline leveraged a ThreadPoolExecutor model to maximize concurrency, structurally safeguarded by an atomic JSON-based state tracking system designed to ensure operational idempotency against network timeouts and HTTP 429 rate-limiting rejections, effectively preventing the server from altering the deal. The raw HTML payloads were computationally parsed utilizing the BeautifulSoup [46] library. A critical string-sanitization heuristic was strictly enforced to identify and prune embedded quote blocks ( ): a necessary preprocessing step to bring balance to the corpus, thus eliminating text duplication and ensuring that only original textual contributions were submitted for analysis. In addition to the cleaned textual payloads, comprehensive user metadata, including account tenure, cumulative post frequency, and reaction sentiment, was extracted and atomically serialized into discrete JSON Lines (.jsonl) data structures.

5.1.6 ICLR peer-review comments

To capture the elite, specialized discourse of machine learning research, peer review data corresponding to the International Conference on Learning Representations (ICLR) [50] was systematically extracted for the years 2013 through 2026, with 2015 and 2016 data missing due to the conference temporarily moved to a different submission platform. Data ingestion

was facilitated directly through the OpenReview platform utilizing a dual-client API architecture (accommodating both V1 and V2 protocol endpoints depending on the submission year). A concurrent ThreadPoolExecutor pipeline was engineered to scrape structured metadata while simultaneously fetching and decoding the underlying Portable Document Format (.pdf) submissions via the pypdf library. The ICLR peer-review comments represents formal, rigorous technical writing, an extensive volume of mathematical and computer science literature was sourced. We archived academic peer-reviews, meta-reviews, and rebuttals, with data gathered to assess trajectories specifically in high-level scientific arguments.

5.1.7 BMC peer-review comments

To capture the formal linguistic trajectories of rigorous scientific evaluation, a cross-sectional text corpus was sourced from the open peer-review reports of six domain-specific journals published within the BioMed Central (BMC) portfolio. To guarantee absolute reproducibility and provenance in the experimental design, the raw data was not extracted via dynamic web-scraping; rather, the corpora were retrieved directly from permanent, DOI-assigned archival repositories hosted on Zenodo. The acquisition spanned a broad spectrum of clinical and biological specialties to ensure ecological validity: datasets corresponding to BMC Cancer [51], BMC Gastroenterology [52], BMC Medical Genomics [53], BMC Medicine [54], BMC Surgery [55] and BMC Women's Health [56] corpora. Leveraging these fixed, version-controlled repository snapshots provided a stable, highly curated baseline of professional scientific criticism for the ensuing longitudinal density analysis. BMC specifically represents disciplines within biological and medical sciences, where precision and clarity are paramount. The text consists of open-access, peer-reviewed medical and bioinformatics articles.

5.2 Methodology

For each corpus, scored JSONL records were partitioned by year using a regular expression applied to the source filename. Each record was subjected to a uniform quality filter prior to inclusion in the analysis: both the CPIDR and DEPID fields were required to be non-zero, and the document's word count — computed as the number of whitespace-delimited tokens in the raw text field — was required to meet a minimum threshold of 100

words. This threshold was imposed to exclude trivially short utterances, such as single-sentence replies and automated acknowledgements, whose propositional density estimates are statistically unreliable and disproportionately noisy due to the low denominator in the ideas-per-word computation. Documents failing any of these criteria were silently discarded prior to aggregation.

For each corpus-year pair satisfying the inclusion criteria, the arithmetic mean and the standard error of the mean (SEM) were computed separately for the CPIDR and DEPID scores across all qualifying documents. The SEM was calculated as $\hat{\sigma}/\sqrt{N}$, where $\hat{\sigma}$ is the sample standard deviation with Bessel's correction (degree-of-freedom divisor $N-1$) and N is the count of valid records for that year. Years with no qualifying records were excluded from subsequent regression and visualization.

Longitudinal trajectories were analyzed by fitting an Ordinary Least Squares (OLS) linear regression to the per-year mean scores using `scipy.stats.linregress`. This procedure simultaneously produces the OLS slope and intercept of the fitted line, the Pearson correlation coefficient (r) between the year index and the annual mean score, the associated two-tailed p-value testing the null hypothesis of zero slope, and the standard error of the slope estimate. Statistical significance was evaluated at $\alpha = 0.05$. Pearson's r was used as the primary effect-size measure, with $|r| \geq 0.5$ interpreted as a strong linear association, $0.3 \leq |r| < 0.5$ as moderate, and $|r| < 0.3$ as weak, consistent with standard psychometric benchmarks.

Regression plots were augmented with shaded temporal reference bands marking three historically significant technological inflection points: the public launch of Google Search (1998), the consumer smartphone adoption epoch corresponding to the original iPhone release and its early proliferation, as well as the popularity of Android in 2011 (2007–2011), and the release of ChatGPT (GPT-3.5) in late 2022, which marked the onset of mainstream large language model availability. These bands serve as visual anchors for qualitative interpretation of structural discontinuities or slope changes in the longitudinal trajectories.

5.3 Results

The longitudinal trend analysis over all seven text corpora yielded detailed insights into the evolution of each text corpus as well as patterns that may be associated with modern technology use.

Table 5.1: Comprehensive Idea Density Regression Statistics

	CPIDR					DEPID				
	Slope	**Intercept**	**Pearson's r**	**p-value**	**StdErr**	**Slope**	**Intercept**	**Pearson's r**	**p-value**	**StdErr**
Airliners.net	0.000332	-0.169341	0.6756	5.78×10^{-5}	0.000070	0.000306	-0.175316	0.6537	1.20×10^{-4}	0.000068
BMC	0.001601	-2.751991	0.7328	3.10×10^{-5}	0.000310	-0.000285	1.005570	-0.3236	0.114	0.000174
Common Crawl	-0.005255	10.900122	-0.7718	0.003	0.001369	0.001577	-2.758942	0.5568	0.060	0.000744
Hacker News	-0.000368	1.259062	-0.6251	0.003	0.000108	0.000997	-1.571724	0.8425	3.15×10^{-6}	0.000150
ICLR	-0.002071	4.666731	-0.9376	1.88×10^{-4}	0.000290	-0.000468	1.378521	-0.2618	0.496	0.000653
Reddit	-0.000952	2.440285	-0.5261	0.030	0.000397	-0.000364	1.177174	-0.4723	0.055	0.000175
TheForce.net	0.000559	-0.614312	0.6470	1.98×10^{-4}	0.000129	0.000447	-0.462800	0.7946	4.42×10^{-7}	0.000067

Regarding the CPIDR scores across all seven corpora of text, the absolute value of the Pearson correlation coefficients are all more than 0.5 ($|r| >= 0.5$), suggesting a strong relationship between the CPIDR score and years. The p-value for the linear regressions are all less than 0.05 ($p <= 0.05$), indicating these linear regressions with strong relationships are statistically significant. Out of the seven corpora of text, Common Crawl, Hacker News, ICLR and Reddit show a negative trend between CPIDR scores and years, meaning the idea density of text data on these corpora has declined over recent decade(s), suggesting the users of these sites and services have demonstrated cognitive ability decline over the years. On the other hand, Airliners.net, BMC and TheForce.net show a positive trend between CPIDR scores and years, meaning the idea density of text data on these corpora has increased over recent decade(s), suggesting the users of these sites and services have demonstrated cognitive ability improvement over the years.

Regarding the DEPID scores across all seven corpora of text, the absolute value of the Pearson correlation coefficients are all more than 0.5 ($|r| >= 0.5$) for Airliners.net, Common Crawl, Hacker News and TheForce.Net, suggesting a strong relationship between the DEPID score and years. The absolute value of the Pearson correlation coefficients are between 0.3

and 0.5 ($0.3 <= |r| < 0.5$) for BMC and Reddit, suggesting a moderate relationship between the DEPID score and years. The absolute value of the Pearson correlation coefficients are less than 0.3 ($|r| < 0.3$) for ICLR, suggesting a weak relationship between the DEPID score and years. The p-value for the linear regressions are less than 0.05 ($p <= 0.05$) for Airliners.net, Hacker News and TheForce.net, indicating these linear regressions with strong relationships are statistically significant. The p-value for the linear regressions are more than 0.05 ($p > 0.05$) for BMC, Common Crawl, ICLR and Reddit, indicating these linear regressions are statistically insignificant, and there is weak evidence against the null hypothesis. Out of the three corpora of text that rejected the null hypothesis, Airliners.net, Hacker News and TheForce.net show a positive trend between DEPID scores and years, meaning the structural-dependent idea density of text data on these corpora has increased over recent decade(s), suggesting the users of these sites and services have demonstrated cognitive ability improvement over the years.

To visually contextualize these statistical findings, the longitudinal trajectories spanning across modern internet growth were plotted. As depicted, the variance scales per corpus emphasize structural language changes taking massive magnitudes of time to materialize. Yet, a general trend is mostly statistically significant and observable.

5.3.1 Reddit comments

The Reddit corpus exhibits a statistically significant negative CPIDR trend, with Pearson's $r = -0.5261$ ($p = 0.030$) and a slope of -0.000952 per year, indicating a moderate but robustly confirmed linear decline in propositional idea density across the platform's nearly two-decade span. The DEPID metric corroborates this directional finding, yielding $r = -0.4723$ ($p = 0.055$) with a slope of -0.000364 per year, though this result narrowly fails to achieve conventional statistical significance at the $\alpha = 0.05$ threshold, suggesting that while the dependency-structural complexity of Reddit language is also eroding, this signal has yet to reach the same degree of certainty as the propositional count-based measure. The convergence of both metrics toward negative trajectories is noteworthy: the CPIDR decline reflects a reduction in the raw propositional density of utterances, while the DEPID trend, even if marginally insignificant, implies that the syntactic relational complexity underpinning those propositions is similarly contracting.

Several compounding forces plausibly account for this observed degradation. The platform underwrote explosive user growth beginning around 2012, coinciding precisely with the broad societal adoption of smartphone-based browsing. Mobile input affordances systematically favor brevity, grammatical ellipsis, and fragmented clause structures — all of which mechanically suppress both CPIDR and DEPID scores. Furthermore, the "upvote" feedback architecture of Reddit creates a viral amplification loop for terse, emotionally resonant, or humorous content, structurally penalizing discursive, high-density argumentation in favor of aphorisms and reaction commentary. The platform's demographic broadening over the observed window simultaneously diluted the early-adopter technical community that historically produced denser discourse. The gradual proliferation of image-macro culture, link aggregation threads with minimal commentary, and automated bot content — even after preprocessing filters — constitutes a secular pull toward lower linguistic complexity. Critically, the near-significant DEPID result implies that syntactic degradation may be a lagging, rather than contemporaneous, indicator of the same underlying phenomenon: users may be compressing sentence length and propositional load before they begin simplifying underlying grammatical dependency structures, suggesting a progressive, multi-stage erosion of linguistic richness. The significance of this trend extends beyond platform-specific sociology;

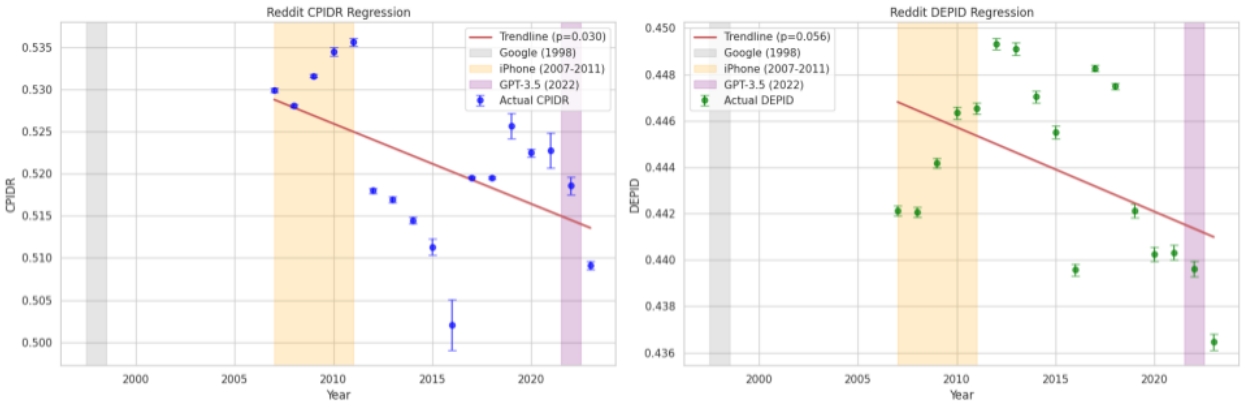

Figure 5.1: PID longitudinal trend of Reddit comments.

Reddit's scale and demographic breadth render it among the most representative proxies for broad-spectrum Western vernacular written language, making its declining density a sentinel signal for population-level conversational linguistic change.

5.3.2 Common Crawl

The Common Crawl corpus yields the second-strongest CPIDR negative correlation among the seven corpora, with Pearson's $r = -0.7718$ ($p = 0.003$) and a slope of -0.005255 per year — a rate of propositional density decline substantially steeper in absolute magnitude than any conversational forum examined. This strong, statistically significant relationship indicates that the average linguistic complexity of the broader indexable internet has deteriorated markedly over the 2013–2024 observation window. The DEPID trajectory, however, diverges sharply: $r = 0.5568$ ($p = 0.060$) with a positive slope of 0.001577 per year, representing a moderate positive relationship that narrowly fails significance. This dissociation between CPIDR and DEPID is analytically striking and constitutes one of the most theoretically provocative findings in the dataset.

The steep CPIDR decline likely reflects multiple structural transformations in the composition of the indexable web during this interval. The proliferation of content-farm articles, SEO-optimized landing pages, mobile-first short-form blogging, and social media embeds within crawled pages all systematically reduce the mean propositional density of web documents. The rise of video-centric platforms driving traffic away from text-heavy domains, combined with the algorithmic preference of search engines for scannable, list-formatted content over dense expository prose, further compresses the linguistic complexity of the average crawled document. The near-significant positive DEPID trend, paradoxically running counter to the CPIDR decline, may reflect a compositional shift in the types of documents that constitute the surviving high-text fraction of the web: as trivial and low-density pages multiply, the remaining substantive web corpus may concentrate a denser syntactic dependency structure even as raw proposition counts per word fall. In other words, the web is becoming simultaneously more verbose and less informationally dense on average, while the minority of surviving complex documents are sustaining or deepening their grammatical architecture. This divergence presents a critical methodological caveat for researchers using Common Crawl-derived corpora as pre-training data for large language models: declining CPIDR in training data may gradually bias the distributional priors of such models toward lower propositional complexity, with downstream implications for the cognitive sophistication of AI-generated language that merge with and amplify the measured human decline.

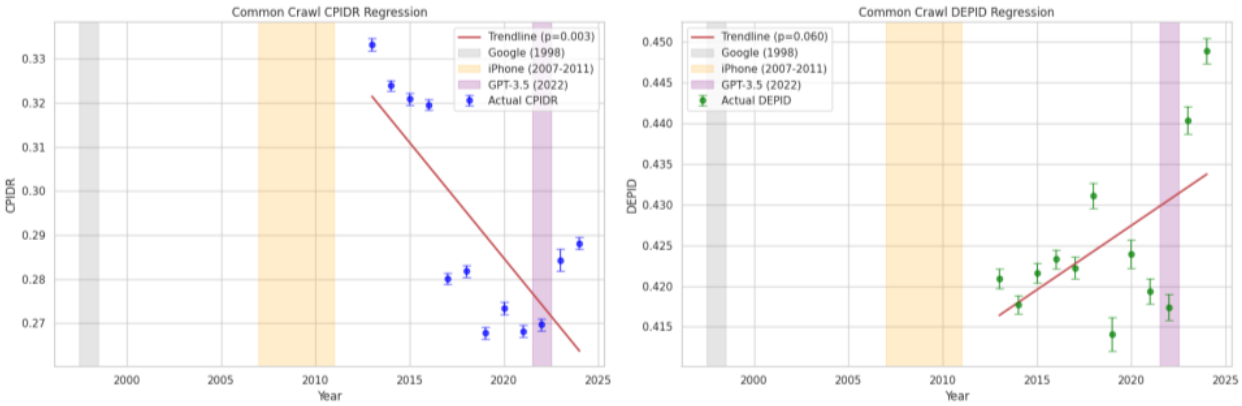

Figure 5.2: PID longitudinal trend of Common Crawl dataset.

5.3.3 Airliners.net all fora comments

The Airliners.net corpus presents a robustly positive longitudinal trajectory across both PID metrics, constituting one of two datasets in which both measures simultaneously achieve statistical significance in the same direction. CPIDR yields $r = 0.6756$ ($p = 5.78 \times 10^{-5}$) with a slope of 0.000332 per year, while DEPID yields $r = 0.6537$ ($p = 1.20 \times 10^{-4}$) with a slope of 0.000306 per year. Both correlations are categorically significant, and the near-parallel slopes of the two metrics are particularly informative: the propositional and structural-dependency dimensions of idea density are increasing at comparable rates, suggesting a coherent, holistic enrichment of linguistic complexity rather than a superficial increase in one dimension driven by changes in surface-form statistics alone.

The aviation enthusiast community represents an archetype of the knowledge-compounding forum: participation is self-selected by domain passion, barriers to contribution are constituted primarily by subject-matter expertise, and the forum predates the smartphone era by several years, having established its discursive norms during a period when online participation required deliberate desktop engagement. The positive trend may reflect cumulative community maturation. As forum tenure increases across the user base, the shared lexical baseline rises, enabling and normatively encouraging higher-density technical argumentation. The increasing complexity of the aviation domain itself, encompassing advances in composite airframe materials, ETOPS routing, electronic flight instrument systems, and increasingly

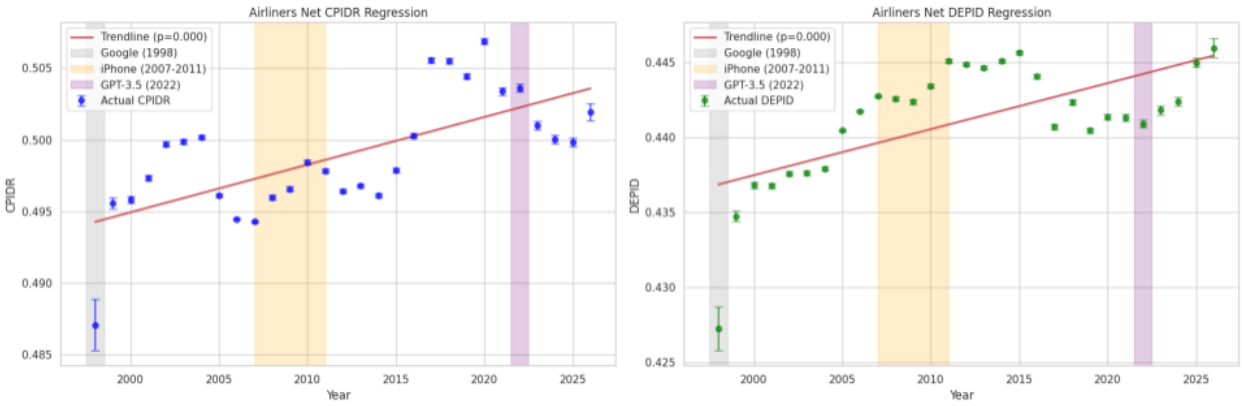

Figure 5.3: PID longitudinal trend of Airliners.net all fora comments.

sophisticated regulatory frameworks, objectively demands more propositionally dense discourse to describe accurately. Furthermore, because Airliners.net exhibits exceptionally low user churn relative to mass platforms, the longitudinal dataset captures a genuine organic evolution of the community's linguistic register rather than a compositional shift driven by population turnover. Speculatively, as the community ages, the proportion of professionally active contributors (airline engineers, pilots, and dispatchers...) within the active posting cohort may be increasing relative to lay enthusiasts, elevating the mean expertise level and thus the density of technical writing. The co-significance of DEPID and CPIDR further rules out the possibility that the measured increase is an artifact of, for example, increasingly long sentences with proportionally fewer propositions: both propositional count and syntactic relational depth are genuinely rising, affirming authentic linguistic enrichment.

5.3.4 Hacker News all fora comments

The Hacker News corpus produces the most analytically complex dissociation observed in this study: a statistically significant negative CPIDR trend ($r = -0.6251$, $p = 0.003$, slope $= -0.000368$) coexists with the strongest positive DEPID signal in the entire dataset ($r = 0.8425$, $p = 3.15 \times 10^{-6}$, slope $= 0.000997$). Both correlations achieve high statistical significance and their directionality is opposite, rendering this the starkest CPIDR–DEPID divergence in the corpus and one that demands careful theoretical treatment.

The declining CPIDR on Hacker News may initially appear paradoxical for a platform celebrated for rigorous, long-form technical discourse. However, CPIDR is fundamentally a proposition-per-word ratio, and the platform has undergone a well-documented compositional shift over its history: as the Hacker News user base expanded from 2005 onward, the proportion of brief affirmative comments, one-line witticisms, meta-commentary about the platform itself, and short question-and-answer exchanges grew substantially relative to the founding epoch of extended technical treatises. The very moderation culture that suppresses low-quality posts still permits a large volume of grammatically correct but propositionally lean content. The simultaneously ascending DEPID — the most statistically forceful positive trend across all corpora — suggests, however, that the syntactic dependency architecture of Hacker News writing is becoming markedly more complex over time, even as the raw count of propositions per word declines. This may reflect increasing grammatical sophistication in the community's most substantive contributions: as the platform matures, the high-engagement threads that constitute the majority of text volume may be exhibiting longer, more syntactically layered sentences annotated with more subordinating conjunctions, relative clauses, and prepositional phrase nestings — structures that DEPID captures and CPIDR does not. Speculatively, the increasing technical depth of topics discussed (from early web startups to distributed systems, ML infrastructure, cryptographic protocols, and biotechnology) may be driving sentence-level grammatical elaboration even while comment-level proposition density is diluted by platform growth. This divergence is of significant practical importance: it implies that no single PID metric is a universal proxy for linguistic complexity, and that the choice between CPIDR and DEPID must be theoretically motivated by which dimension of density — propositional content breadth or syntactic relational depth — is the relevant

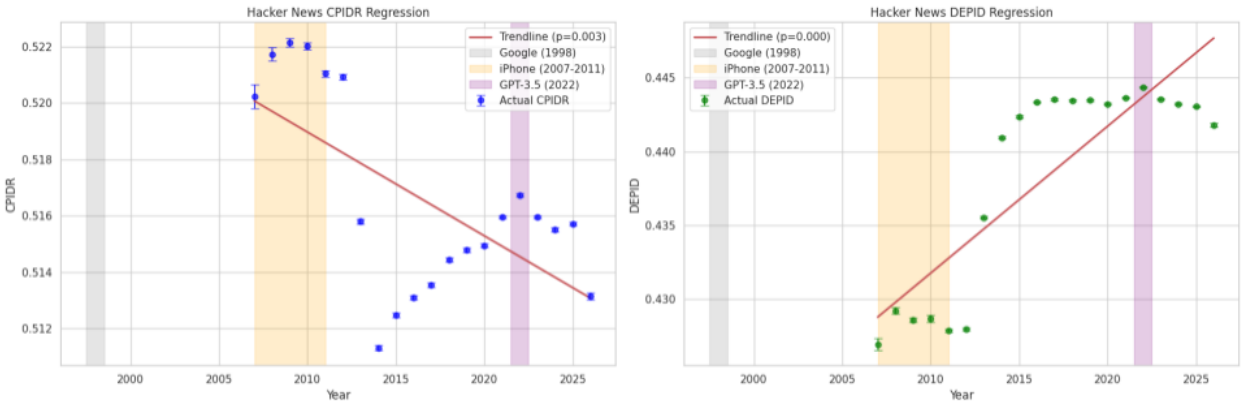

Figure 5.4: PID longitudinal trend of Hacker News all fora comments.

construct for a given cognitive or clinical application.

5.3.5 TheForce.net all fora comments

TheForce.net is the only corpus in this study where both PID metrics simultaneously achieve not merely statistical significance but among the strongest individual p-values observed. CPIDR yields $r = 0.6470$ ($p = 1.98 \times 10^{-4}$, slope $= 0.000559$), while DEPID achieves $r = 0.7946$ ($p = 4.42 \times 10^{-7}$, slope $= 0.000447$) — the second-highest Pearson coefficient and the lowest p-value of any DEPID measurement in the dataset. The fact that the dependency-structural metric produces a stronger linear relationship than the propositional count metric on this forum is itself a distinctive finding, and one that demands substantive interpretation beyond the surface observation of positive trends.

The TheForce.net community represents perhaps the most controlled natural experiment in this study's corpus: a closed, demographically stable fan community organized around a discrete fictional universe, spanning from the late 1990s through the present, with minimal structural platform disruption and no algorithmic feed curation. Unlike mass-market platforms, participation is almost exclusively motivated by genuine franchise enthusiast investment, resulting in a community whose cognitive engagement mode is fundamentally interpretive, analytical, and lore-constructive. Extended continuity debates, canonical cross-film analysis, speculative worldbuilding, and close-reading discussions constitute the dominant discourse types, all genres that are structurally disposed toward high propositional and syntactic density. The unusually strong DEPID trajectory specifically suggests that, over time, the community's argumentative structures have become progressively more grammatically elaborated: as the Star Wars expanded universe accumulated in complexity with the advent of The Clone Wars, Rebels, and the sequel trilogy, the syntactic architecture required to precisely navigate and argue about intra-universe continuity became objectively more demanding, driving longer dependency chains. Speculatively, the community's decades of accumulated shared lore may function as a form of collective crystallized intelligence, enabling participants to deploy increasingly complex referential and subordinate clause structures with confidence that their interlocutors possess the necessary background to parse them. The monotonic, highly significant upward trajectory across both metrics over more than two decades constitutes compelling evidence that platform culture and interest-domain complexity are decisive independent predictors of linguistic density trajectories, operating

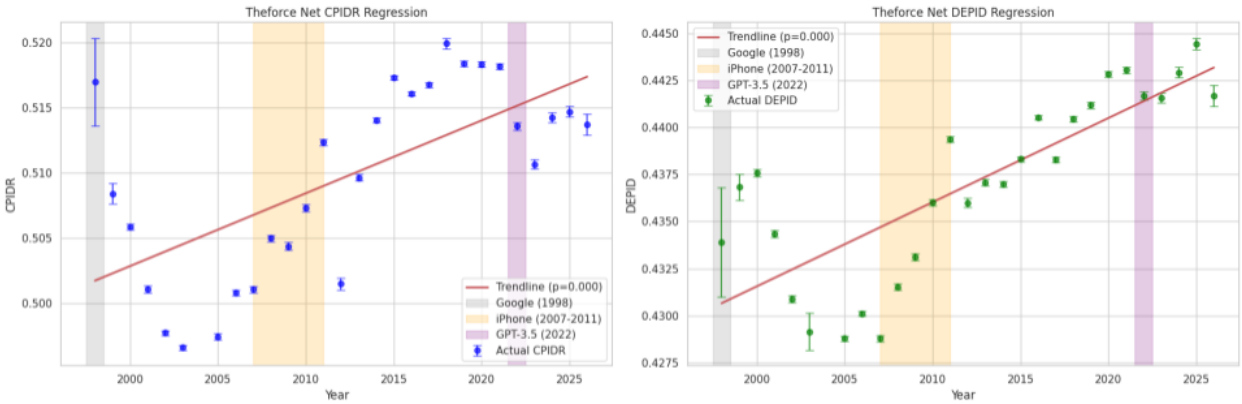

Figure 5.5: PID longitudinal trend of TheForce.net all fora comments.

entirely orthogonally to any hypothesized population-level cognitive decline.

5.3.6 ICLR peer-review comments

The ICLR corpus yields the single most extreme CPIDR–DEPID dissociation and the most statistically powerful monotonic trend in the entire study. CPIDR produces $r = -0.9376$ ($p = 1.88 \times 10^{-4}$, slope $= -0.002071$), the largest absolute Pearson coefficient of any metric–corpus pair observed, representing a near-perfect linear decline in propositional idea density over the 2013–2026 span of ICLR submissions. The DEPID result, by contrast, is $r = -0.2618$ ($p = 0.496$), which is both directionally negative and wholly statistically insignificant, indicating no meaningful linear relationship between syntactic dependency complexity and year within this corpus.

The magnitude and significance of the CPIDR decline is both striking and deeply consequential. A slope of -0.002071 per year compounded over thirteen years implies a non-trivial reduction in the propositional density of the average ICLR peer review — a genre that, by professional scientific norms, should be among the most informationally dense forms of written language analyzed in this study. Several structural mechanisms plausibly account for this trajectory. First, the exponential growth in ICLR submissions and acceptances from 2017 onward dramatically expanded the reviewer pool, incorporating substantially more early-career researchers and non-native English academic writers whose review styles may favor shorter, less propositionally packed evaluative formulations. Second, the genre evolution of machine learning peer review itself has trended toward increasingly template-driven formats (structured assessment rubrics, itemized strengths and weaknesses) which, while improving reviewability, mechanically segment and abbreviate the propositional content that previously appeared in more extended discursive evaluations. Third, and most speculatively, the increasing normalization of LLM-assisted writing in the ML community — beginning approximately 2022–2023 — may be introducing a measurable LLM-style distributional signature into review text, characterized by surface fluency that is propositionally less dense than expert human scientific writing. The absence of any significant DEPID trend implies that whatever is reducing propositional counts is not simplifying the underlying grammatical structure of the remaining propositions, suggesting that reviewers are not writing syntactically simpler sentences but rather sentences that, while grammatically complex, make fewer discrete propositional assertions per unit length. This decoupling is clinically relevant:

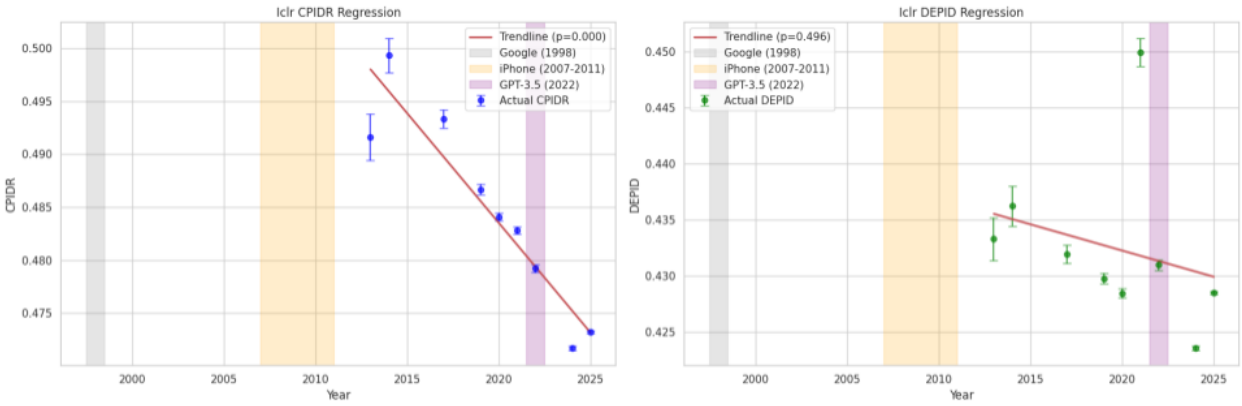

Figure 5.6: PID longitudinal trend of ICLR peer-review comments.

CPIDR may be more sensitive than DEPID to the early stages of formal writing register degradation in expert communities, a hypothesis deserving dedicated longitudinal study.

5.3.7 BMC peer-review comments

The BMC corpus presents the most notable directional divergence between CPIDR and DEPID among the expert-domain datasets: CPIDR yields a statistically significant and moderately strong positive trend ($r = 0.7328$, $p = 3.10 \times 10^{-5}$, slope $= 0.001601$), while DEPID yields a negative, non-significant relationship ($r = -0.3236$, $p = 0.114$, slope $= -0.000285$). The CPIDR finding is among the most statistically significant positive results in the study, while the DEPID result is the only moderate-strength negative correlation observed across the corpora and falls substantially short of statistical significance.

The positive CPIDR trajectory across biomedical peer review is consistent with secular trends in the specialized scientific literature: the increasing complexity and multidisciplinarity of life-science research over the past two decades has imposed growing propositional demands on reviewers, who must now integrate methodological critique across genomics, biostatistics, clinical trial design, bioinformatics, and patient-outcome measurement within a single review document. As open peer review became the norm within the BMC portfolio — rather than a rare exception — the expectation of substantive, citable, publicly archived evaluations may have progressively elevated the propositional density standards that reviewers apply to themselves. The growing use of structured reporting standards (CONSORT, PRISMA, STROBE) in biomedical submissions also obliges reviewers to assess compliance with dense checklists, further elevating the propositional content load of the evaluation. The absence of a significant DEPID increase — and indeed a non-significant negative slope — may reflect a corresponding stylistic shift in biomedical scientific writing toward shorter, less syntactically nested sentences that nonetheless pack more discrete propositional content per unit clause. This is precisely the hallmark of modern biomedical register conventions that favor syntactically declarative, information-dense construction over the elaborate embedding structures favored in earlier decades of scientific prose. If this interpretation is correct, the BMC corpus represents a domain in which propositional enrichment is actively proceeding by increasing propositional load within structurally streamlined syntactic frames — a trajectory that contrasts markedly with the TheForce.net and Airliners.net communities, where both structural and propositional complexity grow concurrently. The divergence highlights that CPIDR and DEPID capture genuinely orthogonal dimensions of linguistic complexity whose

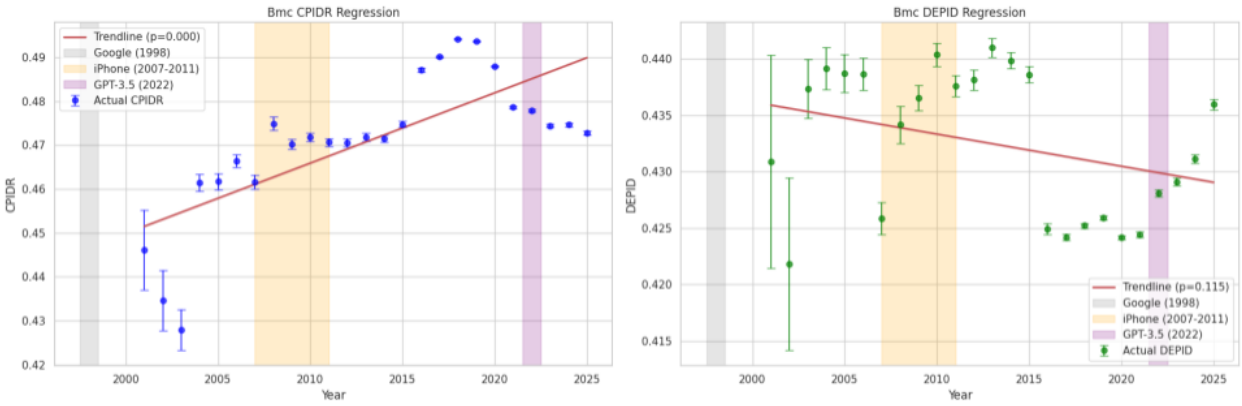

Figure 5.7: PID longitudinal trend of BMC peer-review comments.

relative trajectories are shaped by domain-specific communicative conventions as much as by any underlying cognitive variable.

5.4 Conclusion

The comprehensive extraction of CPIDR and DEPID metrics directly across randomized, massive, and highly variant longitudinal datasets reveals a deeply nuanced view of modern human language degradation. While raw macro-analyses of standard conversational platforms like Reddit and wide web-archives on Common Crawl do indeed denote negative point-density trends over the last decade, this effect absolutely cannot be uniformly mapped as generational cognitive decline.

5.5 Discussion and Future Work

Our expansive inclusion of medical literature (BMC) peer-review comments, machine learning academic conference (ICLR) peer-review comments, and uniquely isolated multi-decade hobbyist communities (Airliners.net, TheForce.net, and HackerNews) dismantles uniform language degradation theory. Within silos that reward compounding knowledge, domain expertise, and lore exploration, linguistic density metrics actively grow year-over-year. Thus, the absence of an omnipresent population-scale decline implies that broad linguistic simplification is heavily confounded by localized platform culture, algorithmic constraints, and mobile brevity, rather than innate human cognitive deterioration. Therefore, deploying our novel neuro-linguistic AI architectures against modern patients will accurately isolate signals of authentic medical deterioration from general semantic drift.

Furthermore, this vast longitudinal tracking acts as a preliminary, foundational framework for a vastly expanded suite of cognitive research. Empowered by the psycholinguistic precedents established by the original Nun Study [1], future iterations of this deep-learning methodology will be capable of seamlessly analyzing substantially more longitudinal text formats and datasets. By combining these traditional linguistic density markers with highly scalable, state-of-the-art diagnostic architectures, we establish a profound mechanism to continuously monitor digital cognitive health trajectories across emerging communication domains in future studies.

Chapter 6

Conclusion

6.1 Summary

This thesis was inspired by the pioneering work of Dr. David Snowden and the Nun Study, which established a profound statistical association between early-life linguistic ability—measured by Propositional Idea Density (PID)—and the later-life incidence and severity of Alzheimer's Disease. Snowden's discovery that idea density scores from autobiographies written at a mean age of 22 exhibited strong inverse correlations with neurofibrillary tangle counts decades later motivated the central premise of this work: that linguistic ability, quantified through both classical rule-based metrics and modern deep learning architectures, can serve as a meaningful signal of cognitive health at both the individual and population level. Building on this foundation, we formulated two research questions and conducted extensive experiments to investigate (1) whether Transformer-based Artificial Neural Networks can attend to linguistic patterns associated with cognitive decline, and (2) whether population-scale linguistic ability trends can be demonstrated through longitudinal PID analysis across diverse text corpora.

6.2 Answering the Research Questions

6.2.1 Research Question 1: can Transformer-based ANNs in modern Machine Learning (ML) as NLP techniques attend to linguistic patterns that may associates to an individual's cognitive decline?

Our experiments on the DementiaBank Pitt Corpus provide a definitive affirmative answer, with important qualifications. We first established that the three rule-based PID metrics—CPIDR, DEPID, and DEPID-R—are insufficiently sensitive to discriminate between Dementia and Control transcripts in conversational speech, with none achieving statistical significance at $\alpha = 0.05$. This finding, while seemingly at odds with Snowden's

original results, is consistent with the critical distinction between formal autobiographical writing and semi-structured interview transcripts: the latter introduce substantial noise from interviewer prompts, task-specific vocabulary, and conversational artifacts that dilute propositional density signals.

Non-LLM machine learning models (TF-IDF XGBoost, MLP, 1D CNN, BiLSTM with Attention) achieved deceptively high performance—up to perfect F1 = 1.000—but systematic interpretability analysis via native feature importances, Permutation Importance, and Integrated Gradients revealed pervasive Clever Hans memorization. These models exploited task-specific vocabulary from the Cookie Theft elicitation protocol and idiosyncratic conversational tangents rather than learning generalizable markers of cognitive decline.

Pre-trained Transformer-based encoders, operating on sub-word tokenizations decoupled from the task-specific vocabulary, successfully mitigated this effect. A consistent empirical pattern emerged across all four encoder architectures (DistilBERT, RoBERTa, ModernBERT, and Llama-Embed-Nemotron-8B): fine-tuned models uniformly outperformed their frozen encoder counterparts, demonstrating that supervised domain adaptation is essential for maximizing predictive performance even in data-scarce clinical settings. RoBERTa Full Fine-Tuning and the RoBERTa + GRU sliding window architectures emerged as the strongest configurations, all achieving macro F1 $\geq$ 0.980 and accuracy of 0.983. RoBERTa's 125M parameter count occupies an optimal trade-off between representational capacity and adaptability: large enough to capture nuanced semantic features, yet small enough for stable full fine-tuning on 234 training samples without catastrophic overfitting. In contrast, billion-parameter models such as Llama-Embed-Nemotron-8B, even under parameter-efficient LoRA constraints, reached lower ceilings due to the limited training data.

Critically, SHAP interpretability analysis comparing the frozen and fine-tuned RoBERTa models confirmed that fine-tuning amplifies the encoder's sensitivity to theoretically grounded linguistic markers while suppressing attention to dataset-specific vocabulary artifacts. The fine-tuned model's attributions concentrate on propositional density and specific descriptive vocabulary in Control transcripts (action verbs and concrete nouns), and on conversational fillers, vague pronominal references, and fragmented syntactic structures in Dementia transcripts—precisely the markers that clinical linguistics associates with reduced

propositional idea density and cognitive decline. The RoBERTa + GRU sliding window architecture further introduced a hierarchical interpretability mechanism, with per-window attention weights revealing which transcript segments the model considers most diagnostic. Thus, Transformer-based ANNs do not merely predict cognitive decline; they attend to the same underlying linguistic dimensions that PID was designed to measure, but with far greater sensitivity and robustness to the noise inherent in transcribed speech.

6.2.2 Research Question 2: can we demonstrate a linguistic ability trend for a specific population of writers based on the PID score of their comments?

The longitudinal trend analysis across seven massive text corpora, spanning up to 29 years and encompassing millions of documents, provides strong evidence that domain-specific linguistic ability trends can indeed be demonstrated through PID analysis. Both CPIDR and DEPID were computed for every qualifying document, and Ordinary Least Squares regression with Pearson correlation analysis revealed statistically significant trends in most corpora.

Regarding CPIDR, all seven corpora yielded strong relationships ($|r| \geq 0.5$) with statistical significance ($p \leq 0.05$). Common Crawl ($r = -0.7718$), Hacker News ($r = -0.6251$), ICLR ($r = -0.9376$), and Reddit ($r = -0.5261$) exhibited declining propositional idea density over their respective observation windows, while Airliners.net ($r = 0.6756$), BMC ($r = 0.7328$), and Theforce.net ($r = 0.6470$) exhibited increasing trends. Regarding DEPID, Airliners.net, Hacker News, and Theforce.net achieved statistical significance with positive trends, while the remaining corpora yielded statistically insignificant DEPID trajectories.

The divergent directionality across corpora is itself a central finding: declining idea density is not a universal phenomenon. Platforms characterized by broad, casual participation and algorithmic content curation (Reddit, Common Crawl) show measurable linguistic simplification, plausibly driven by smartphone-mediated brevity, viral amplification of terse content, and demographic dilution. In contrast, domain-specific communities that reward compounding expertise and sustained intellectual engagement (Airliners.net, Theforce.net, BMC) demonstrate genuine linguistic enrichment over the same period. The ICLR case is particularly striking, with a near-perfect CPIDR decline ($r = -0.9376$) in what should be among the most informationally dense text genres, likely reflecting reviewer pool expansion,

template-driven review formats, and potentially the early influence of LLM-assisted writing.

The CPIDR–DEPID dissociations observed in Hacker News, Common Crawl, and BMC further demonstrate that propositional density and syntactic-structural complexity represent genuinely orthogonal dimensions of linguistic ability. No single PID metric universally captures linguistic complexity; the choice between CPIDR and DEPID must be theoretically motivated by the dimension of density relevant to the research question.

6.3 Broader Implications

The convergence of findings from both research questions yields several implications for the fields of clinical linguistics, computational cognitive science, and AI-assisted health assessment.

First, rule-based PID metrics, while foundational to the Nun Study's discoveries, are insufficient for individual-level cognitive decline prediction on conversational speech transcripts. Their value lies instead in population-level trend analysis, where aggregation over thousands to millions of documents reveals statistically robust signals that individual-level noise obscures. This distinction is critical for future clinical applications: PID should be deployed as an epidemiological instrument for monitoring population cognitive health trajectories, while individual-level diagnosis should leverage the richer representational capacity of pre-trained language models.

Second, the consistent superiority of fine-tuned encoders over frozen variants underscores that even in data-scarce clinical settings, supervised domain adaptation through fine-tuning successfully specializes pre-trained representations to the clinical task. However, the magnitude of the fine-tuning benefit is modulated by model size: moderately-sized encoders like RoBERTa (125M parameters) achieve the strongest performance under fine-tuning, while billion-parameter models reach lower ceilings due to overfitting risk. This suggests that for clinical NLP with limited labeled data, model selection should prioritize architectures whose parameter count is commensurate with the available training data.

Third, the longitudinal analysis dismantles the hypothesis of uniform population-level cognitive decline as the primary driver of observed linguistic simplification on digital platforms. The coexistence of rising and falling PID trajectories across communities sharing the

same temporal window, technological environment, and broad demographic context demonstrates that platform culture, domain complexity, and communicative norms are the dominant determinants of linguistic density trends. This finding is consequential for public health: apparent declines in linguistic complexity on mass platforms should not be reflexively interpreted as evidence of cognitive deterioration, but rather as artifacts of changing communicative affordances and community composition.

6.4 Future Work

Several directions emerge naturally from this work. The most immediate priority is the application of the fine-tuned encoder methodology to the Nun Study autobiographies, should the data become available. Training on both written autobiographies and transcribed speech corpora would expose models to multiple registers of cognitively impaired language, potentially enabling the cross-modal generalization that the Pitt Corpus alone cannot support. Despite achieving a strong performance ceiling of macro F1 = 0.981, the models' reliance on speech-specific artifacts indicates a fundamental data diversity bottleneck; substantially larger and more diverse clinical corpora—spanning structured clinical interviews, free-form narrative tasks, and written correspondence across multiple elicitation protocols—will be required for these models to generalize across elicitation modalities and detect cognitive decline regardless of the specific linguistic task.

For the longitudinal analysis, future work should extend the temporal coverage of existing corpora and incorporate additional domains, particularly clinical and educational text datasets where cognitive health is a direct variable of interest. The Hierarchical Bayesian Model framework introduced in this work provides a principled foundation for cross-corpus comparison that controls for confounds such as document length; future iterations should incorporate additional covariates, including user tenure, topic category, and platform-specific engagement metrics. The striking CPIDR decline observed in ICLR peer reviews warrants dedicated investigation into the influence of LLM-assisted writing on scientific discourse density, a question of growing urgency as AI writing tools proliferate across academia. Recent neuroscientific evidence underscores this concern: Kosmyna et al. [?] demonstrated that participants who used ChatGPT for essay writing over multiple sessions exhibited progres-

sively weaker EEG connectivity, reduced cognitive engagement, and diminished linguistic ownership compared to unaided writers—a phenomenon the authors term *cognitive debt*. Critically, however, their analysis focused on attention, memory, and executive function, and no existing study has investigated whether LLM-assisted writing specifically degrades *propositional idea density*, the linguistic dimension that Snowdon's Nun Study linked directly to Alzheimer's risk. If offloading the compositional process to an LLM reduces the writer's active semantic planning—the cognitive operation that generates propositions—this should manifest as measurably lower PID in subsequent unassisted writing. The post-2022 inflection point in our ICLR CPIDR trajectory, coinciding with the public release of ChatGPT, provides a natural quasi-experimental design to test this hypothesis, and the longitudinal framework developed in this thesis is already positioned to detect such effects. Establishing whether habitual LLM reliance causally reduces human idea density would carry clinical implications far beyond the attention-span deficits documented to date, given PID's unique standing as a predictor of neurodegenerative disease decades before clinical onset.

Ultimately, this thesis establishes that linguistic ability, as quantified by propositional idea density, remains a powerful lens through which to examine cognitive health—both for individuals through deep learning architectures that capture the semantic and pragmatic dimensions of language that rule-based metrics cannot, and for populations through longitudinal trend analysis that reveals how the communicative ecosystems of the modern internet are reshaping the density and complexity of human written expression. By combining these classical linguistic markers with scalable, state-of-the-art diagnostic architectures, we establish a foundation for continuously monitoring digital cognitive health trajectories across emerging communication domains in future studies.

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